

Adequacy Score Between Declared and Real Urgency in a Digital Smart Green Supply Chain

R&D BST, Digital Smart Green Supply Chain: Planning Score



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2026
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Remerciements

Je tiens à remercier l'ensemble des personnes qui ont accompagné le déroulement de cette opération de R&D au sein de la Direction Innovation Numérique d'ALTEN.

Je remercie tout particulièrement **Émilienne Lardy**, **Selma Ferhat** et **Sébastien Perthuisot**. Émilienne et Selma ont accompli un travail remarquable dans mon encadrement et mon suivi scientifique : leur exigence, leur disponibilité et la qualité de leurs relectures ont directement fait progresser la rigueur de ce mémoire. Sébastien, Directeur de Programme de Recherche et Expert Qualité & Supply Chain à la DIN, m'a apporté son encadrement industriel et sa confiance sur l'orientation scientifique du sujet.

Je remercie mes encadrants académiques, **David Gomez**, Professeur en Informatique à l'ESTIA, et **Peter Sherar**, du Centre for Computational Engineering Sciences de Cranfield University, pour leurs conseils méthodologiques et le cadrage académique apporté à ce mémoire.

Enfin, je remercie l'ensemble de l'équipe de la Direction Innovation Numérique, ainsi que les autres stagiaires du département, pour leur accueil et les échanges qui ont nourri ce travail tout au long du stage.

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Abstract

Modern supply chains operating in Physical Internet networks face a fundamental signal-quality gap: the urgency declared by operators often diverges from the physically computable real urgency. This leads to over-triage (resource waste) and under-triage (hidden rupture risk). In Supply Chain 4.0 contexts marked by volatility and multi-tier dependencies, no existing digital twin architecture fully corrects this cognitive bias while propagating it consistently across a supplier network.

This report summarises a R&D internship that proposes a cognitive adequacy score $A(t) \in [0, 100]$ to measure and correct the gap between declared operator urgency and a multi-block, DAG-propagated digital twin real urgency $U_r(t)$. The approach combines pairwise elicitation (AHP, Fuzzy Best-Worst Method) for declared urgency, a noisy-OR aggregation of six operational KPI blocks for real urgency, PROMETHEE II outranking and systematic shock simulation for network-wide criticality, and a calibrated event engine for keeping the underlying KPI state current. The framework is implemented in Python and operated as a persistent, multi-tenant digital twin exercised through a weekly-cadence serious game, with a dedicated calibration service comparing the adequacy signal against recorded operational outcomes.

Résumé

Les chaînes logistiques modernes opérant dans les réseaux de l'Internet Physique font face à un écart fondamental de qualité de signal : l'urgence déclarée par les opérateurs diverge souvent de l'urgence réelle physiquement calculable. Il en résulte du sur-triage (gaspillage de ressources) et du sous-triage (risque de rupture caché). Dans des contextes Supply Chain 4.0 marqués par la volatilité et les dépendances multi-rangs, aucune architecture de jumeau numérique existante ne corrige pleinement ce biais cognitif tout en le propageant de manière cohérente sur un réseau de fournisseurs.

Ce mémoire synthétise un stage de R&D qui propose un score d'adéquation cognitive $A(t) \in [0, 100]$ pour mesurer et corriger l'écart entre l'urgence déclarée par l'opérateur et une urgence réelle $U_r(t)$ multi-blocs, calculée par le jumeau numérique et propagée sur le graphe orienté acyclique du réseau. L'approche combine une élicitation par comparaisons par paires (AHP, Fuzzy Best-Worst Method) pour l'urgence déclarée, une agrégation noisy-OR de six blocs de KPI opérationnels pour l'urgence réelle, le surclassement PROMETHEE II et une simulation systématique de chocs pour la criticité réseau, ainsi qu'un moteur d'événements calibrés pour maintenir à jour l'état des KPI sous-jacents. Le cadre est implémenté en Python et opéré comme un jumeau numérique persistant et multi-tenant, exercé à travers un serious game à cadence hebdomadaire, avec un service de calibration dédié comparant le signal d'adéquation aux résultats opérationnels enregistrés.

1 R&D Operation Context

1.1 Internal Research Context

1.1.1 Smart Green Supply Chain Programme

The Direction Innovation Numérique (DIN) of ALTEN conducts applied R&D at the intersection of digital twin technology, logistics optimisation, and sustainable transport.

The Smart Green Supply Chain programme (Ben Zakour, 2024) articulates three complementary pillars:

- **Lean:** Minimisation of waste (cost, delay, over-stock) through pull-flow logic and real-time scheduling.
- **Green:** Minimisation of carbon footprint per transport leg, aligned with the Global Logistics Emissions Council (GLEC) Framework v3.2 (Smart Freight Centre, 2023) and ISO 14083:2023 (International Organization for Standardization, 2023).
- **Predictive:** Anticipation of disruptions through probabilistic modelling and machine learning, consistent with the digital supply chain twin approach (Ivanov; Dolgui, 2020).

This programme places ALTEN-DIN within the broader Industry 4.0 movement (Kagermann et al., 2013), which uses cyber-physical systems, Internet of Things (IoT), and digital twins to transform reactive supply chains into adaptive, self-organising networks.

The Physical Internet framework (Montreuil, 2011), proposing open, mutualized logistics networks analogous to internet routing protocols, provides the theoretical backdrop for the supply chain simulation environment described below.

1.1.2 Project: DISCO Digital Twin

DISCO (Supply Chain Illustrator) is a serious game and digital twin developed by the DIN to simulate a complete supply chain for a fictitious satellite product. It involves 22 component suppliers in domains such as structural assemblies, optical systems, and embedded electronics (Baxas, 2025). Players assume the role of supply chain buyers and make decisions about supplier selection, transport routes, and factory allocation. The DISCO technology stack is summarised in Table 2.

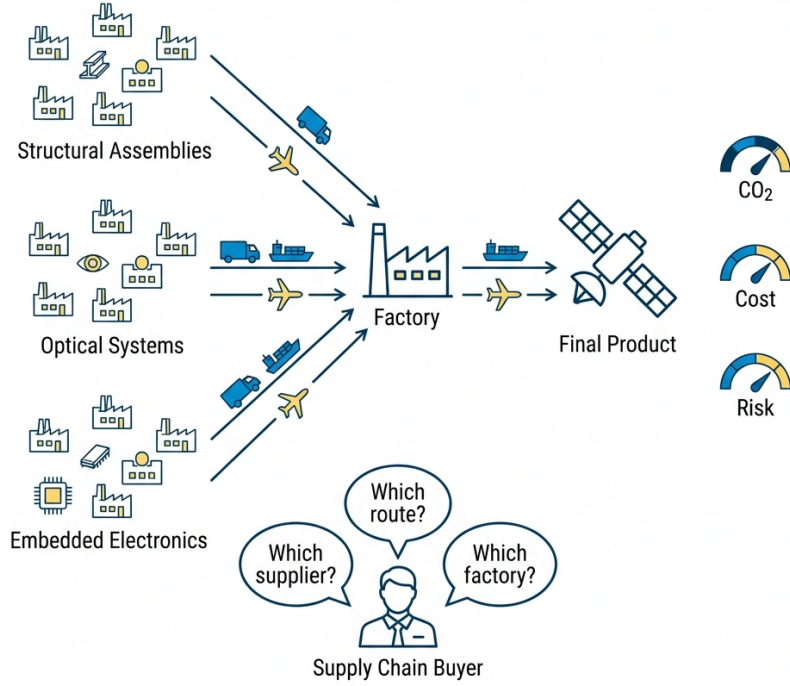


Figure 1: The DISCO serious game: 22 suppliers, transport decisions, and the three scored dimensions.

The scoring engine applies Min-Max normalisation over the values of the current game session:

$$\text{ScoreCO}_2 = 1 + 9 \times \frac{\text{realCO}_2 - \text{co}_{2,\min}}{\text{co}_{2,\max} - \text{co}_{2,\min}}, \quad (1)$$

The operational variables of the CO₂ scoring model are described in Table 1.

Table 1: Component definitions for the CO₂ scoring equation.

Component	Type	Description
ScoreCO ₂	Dependent variable	Normalized greenhouse gas score on a [1, 10] scale
realCO ₂	Input variable	Measured carbon emissions on the transport leg
co _{2,min}	System parameter	Minimum historical or target emissions threshold
co _{2,max}	System parameter	Maximum historical or target emissions threshold

with analogous formulae for cost and risk. The scale ranges from 1 to 10, where 1 corresponds to the best possible performance and 10 to the worst.

To support the demonstration and continuous improvement of mature supply chain capabilities, the DISCO digital twin incorporates several specialized modules developed in prior research cycles: The routing module (Baulain et al., 2024), the risk scorer (Kabouche, 2025), the CO₂ quantification methodology (Cordova, 2024), and the software integration framework (Auclin, 2025).

Table 2: DISCO technology stack and key parameters.

Component	Technology / Value
Backend API	Spring Boot + Java 21
Database	PostgreSQL (Azure)
Frontend	Angular (localhost:4200)
REST API (test)	https://test-disco.altenlabs.net/back/
Authentication	Player key / Azure AD SSO OAuth2
Scores computed	CO ₂ , Cost, Risk per supplier/itinerary [1–10]

1.1.3 Capitalisation of Prior ALTEN-DIN Research

This R&D operation is situated within a body of prior DIN research conducted under the Smart Green Digital Supply Chain programme. These works form the organisational and thematic context of the present operation; the system delivered here (SupplyScore, Section 4) is a standalone Python application and does not integrate with these modules yet. Their coupling is a stated perspective of the programme, not a delivered capability.

Baxas (Baxas, 2025) defines the DISCO software architecture, including the Time To Service (TTS)/Time To Recovery (TTR) formalisation.

Baulain and Dubarry (Baulain et al., 2024) developed the Physical Internet routing module of DISCO, providing transit times via the HERE Maps API (Vincenty distances, mandatory 4h30 Heavy Goods Vehicle (HGV) pause compliance) and the Itinerary Performance Index (Itinerary Performance Index (IPG)) normalised in [0, 1].

Kabouche (Kabouche, 2025) developed the supplier risk scorer combining delay history, geographic vulnerability, and financial fragility into a [1, 10] scale.

Cordova (Cordova, 2024) developed the CO₂ quantification module per route using the GLEC/VECTO methodology with log-normalised output $\hat{x}_{\text{CO}_2}$.

Mondelice (Mondelice, 2025) developed the Transport Management System interface providing real-time transit-time extraction.

1.2 Document Structure

This document is structured as follows. Section 2 formalises the scientific problem, states the operation's objectives, research questions and hypotheses, and outlines the primary research challenges. Section 3 reviews the state of the art across thematic subsections. Section 4 details the R&D iterations, the methodology, and the serious-game validation campaign. Finally, Section 5 states the documented limitations of the delivered framework and closes with a personal assessment.

2 Scientific Problem and Obstacles to Overcome

2.1 Relevant Context and Motivating Observations

Modern supply chain operators face simultaneous multi-dimensional assessments (cost, deadline, risk, carbon) where most monitoring and planning tools process one criterion at a time (Pinedo, 2012; Chakraborty et al., 2020), which is precisely why practitioners fall back on judgment-based prioritisation.

This work addresses the layer that comes *before* any planning or scheduling decision, and which the classical literature has not fully investigated: the urgency signal itself is systematically biased at the source and must be measured and corrected before it can safely inform any downstream decision (M. Zhu et al., 2018; Rusou et al., 2020).

In a Physical Internet network (Montreuil, 2011), where logistics capacity is mutualized across actors, a biased urgency signal propagates and amplifies systemic misallocations. Over-triage (declaring an urgency higher than the physical situation warrants) starves other network tenants, whereas under-triage (declaring an urgency lower than warranted) creates invisible stock rupture risk that cascades downstream (Li et al., 2020; Ivanov; Dolgui, 2020).

While digital twin architectures and real-time decision support systems have demonstrated strong capabilities in operational monitoring, disruption detection, resource reallocation, and order synchronisation (Ding et al., 2023; Leung et al., 2022; Hrušovský et al., 2021; Karam et al., 2022; X. Zhu et al., 2025), they do so primarily by computing the physical operational state of the system. To the best of the present review, existing systems do not appear to explicitly model the gap between a declared client urgency and a computed operational urgency.

The core scientific novelty of this R&D operation therefore lies in the formal comparison between these declared and computed urgencies, propagated coherently across a multi-tier network. This missing integration motivates the introduction of an urgency adequacy layer in digital operations architectures.

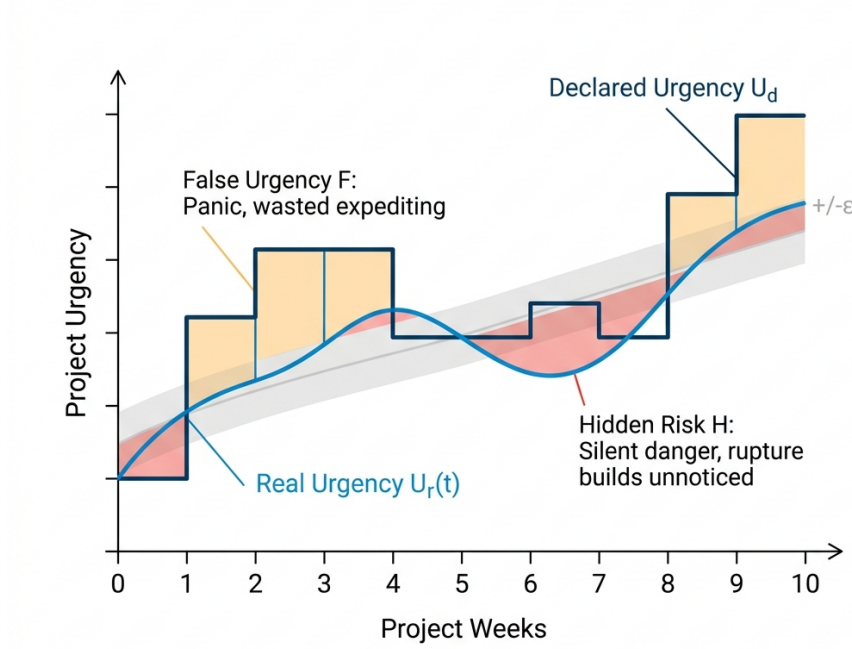


Figure 2: Conceptual view of the declared vs. real urgency gap and its two pathologies.

2.2 Formal Problem Statement

Let $U_d \in [0, 1]$ denote the declared urgency and $U_r(t) \in [0, 1]$ the real urgency computed from operational data in the digital twin. The core variables and domains are summarized in Table 3.

Table 3: Core variables and domains.

Symbol	Meaning	Domain
t	Current time (project hours, ISO week granularity)	$\mathbb{R}$
$U_r(t)$	Physically/operationally computable real urgency	$[0, 1]$
U_d	Operator-declared urgency	$[0, 1]$
ε	Tolerance threshold (cognitive noise band)	$[0, 1]$

The mismatch between both signals is defined as

$$\Delta U(t) = U_d - U_r(t) \in [-1, 1]. \quad (2)$$

The components of this mismatch equation are explained in Table 4.

Table 4: Component definitions for the urgency discrepancy equation.

Component	Type	Description
$\Delta U(t)$	Dependent variable	Raw urgency mismatch signal at time t
U_d	Input variable	Urgency level declared subjectively by the human operator
$U_r(t)$	Dynamic variable	Urgency level computed operationally by the digital twin

A tolerance band $\varepsilon \in [0, 1]$ is introduced so that small deviations are not treated as operationally meaningful. In this setting, over-declaration corresponds to $\Delta U(t) > \varepsilon$, under-declaration to $\Delta U(t) < -\varepsilon$, and approximate agreement to $|\Delta U(t)| \leq \varepsilon$. Potential consequences of these mismatch conditions are summarized in Table 5.

Table 5: Urgency pathologies and potential consequences.

Condition	Interpretation	Potential consequence
$\Delta U(t) > \varepsilon$	Over-declaration of urgency (false urgency)	Excess expediting, avoidable cost, possible delay for other flows
$\Delta U(t) < -\varepsilon$	Under-declaration of urgency (hidden risk)	Hidden lateness or stockout risk, delayed intervention, downstream disruption
$ \Delta U(t) \leq \varepsilon$	Acceptable alignment	No material urgency mismatch

An asymmetric treatment of mismatch may be introduced through weights such as $\lambda_{\text{under}} \geq \lambda_{\text{over}}$ when the application context treats under-prioritisation as more harmful than over-prioritisation. This is motivated by operations literature showing that decision errors can generate unequal downstream consequences (Kim et al., 2020; DiwasSingh et al., 2020), and by scheduling literature that models differentiated tardiness penalties rather than uniform delay costs (Baker et al., 1982; Chauhan et al., 2022). However, the degree of asymmetry is treated here as a design and calibration choice to be validated in the target use case, rather than as a universal property.

2.3 Operation Objectives and Research Questions

The R&D operation is entitled “*Adequacy Score Between Declared and Real Urgency in a Smart Green Supply Chain Digital Twin*”. Its three operational objectives are:

- O1.** Define a multi-block, DAG-propagated real urgency $U_r(t)$ and a cognitive adequacy score $A(t) \in [0, 100]$ to measure the gap between the human planner's declared urgency U_d and the digital twin's operational state.
- O2.** Develop and implement a network propagation and criticality-driven prioritisation layer, combining PROMETHEE II outranking with systematic shock simulation, to identify structurally critical nodes.
- O3.** Validate the adequacy-driven prioritisation logic empirically, through worked scenarios and a calibration service comparing the hidden-risk signal against recorded operational outcomes.

Four research questions structure the scientific approach to these objectives:

- RQ1. Multi-block real urgency ($U_r(t)$):** How can a physically and operationally grounded real urgency $U_r(t)$ be computed dynamically in a multi-tenant digital twin from heterogeneous KPI blocks (temporal, capacity, performance, risk, cost, environmental), and propagated coherently across a multi-tier supply chain graph?
- RQ2. Cognitive adequacy formulation ($A(t)$):** How should an adequacy score $A(t)$ be formulated to measure the alignment between operator declared urgency U_d and computed real urgency $U_r(t)$, while asymmetrically penalizing under-triage vs. over-triage, and can this scoring demonstrate predictive validity against real operational outcomes?
- RQ3. Network prioritisation and criticality:** How can pairwise elicitation methods (AHP, Fuzzy Best-Worst Method) be combined with outranking aggregation (PROMETHEE II) and systematic shock simulation to identify which nodes of a supply network are structurally critical and warrant prioritised attention?
- RQ4. Operational validation within a digital twin:** How can this cognitive-physical framework be operationalised and empirically validated within a persistent, multi-tenant digital twin exercised as a weekly-cadence serious game, and what predictive validity does it demonstrate against recorded operational incidents?

2.4 Research Hypotheses

These hypotheses apply to the entire study. They are formulated here, once, so that the evaluation criteria are fixed before any method is introduced, and they match the pre-registered hypothesis register of the validation campaign described in Section 4.8. Each is falsifiable: it states an observable outcome, with a numeric threshold, that the delivered system either does or does not exhibit.

- H1 (Cognitive latency):** Declared urgency follows computed real urgency with a delay of at least one weekly cycle on the majority of nodes: operators

register a degradation after the physical signal, not with it. Test: the lagged correlation between U_d and U_r peaks at a lag of one week or more for at least five of the eight campaign nodes.

H2 (Early detection of hidden risk): The hidden-risk signal $H = \max(U_r - U_d, 0)$ exceeds 0.3 at the crisis-origin node several weeks before the rupture becomes publicly visible.

H3 (Structural criticality): The systematic shock simulation ranks the true crisis origin first, by impact on the final customer, among deep-tier suppliers, on at least 80% of the informative (non-saturated) weekly rounds.

H4 (Predictive validity): At threshold $H \geq 0.5$ with a four-week horizon, the hidden-risk signal predicts recorded negative outcomes (missed milestone, critical event, node abandonment) with precision and recall both above 0.5.

H5 (Urgency contagion): On weeks when crisis news becomes public, operators of nodes that are not physically affected still raise their declared urgency, reproducing the mere-urgency effect (M. Zhu et al., 2018) at network scale.

H6 (Perception hysteresis): Once the physical situation improves, declared urgency decays at less than half the rate of real urgency: panic outlives its cause.

If H1 and H5 fail, operators need no debiasing and the adequacy layer is redundant. If H2 and H4 fail, the hidden-risk signal is a descriptive curiosity rather than a decision aid. If H3 fails, network prioritisation cannot be trusted operationally. If H6 fails, over-declaration is self-correcting and the asymmetric penalty needs recalibration.

Model Assumptions. Distinct from these testable hypotheses, the framework rests on a set of stated modelling assumptions, each verifiable in the delivered code: the network is a directed acyclic graph (no feedback loops, linear topological sort); reviews follow a strict ISO weekly cycle; declared urgency is damped as it descends toward suppliers while real risk is amplified as it ascends toward the customer, each scaled by a per-arc coefficient; the six KPI blocks are treated as independent to permit a probabilistic aggregation; task statuses drive explicit override rules; and the cognitive loss is deliberately asymmetric, with under-declaration priced at roughly twice the cost of panic. These assumptions are design commitments, not empirical claims; the limitations they induce are catalogued in Section 5.1.

2.5 Technical and Scientific Challenges

To evaluate the validity of the proposed framework, its performance is investigated against five scientific challenges identified in contemporary literature. Each challenge is stated here as an open problem; the design choices made to overcome them are in Section 4.

Challenge 1: Justifying the Urgency Aggregation

Real urgency can be modeled from a single deadline-proximity signal or from an aggregation of heterogeneous risk blocks (slack, capacity saturation, performance, failure exposure, cost drift, emissions overshoot). A single-block, purely time-driven curve is analytically convenient but discards information available in a live digital twin. Conversely, any multi-block aggregation rule must be theoretically justified against simpler linear or averaging alternatives (K. Chen et al., 1992; K. Chen et al., 1996; J. Chen et al., 2025; Mills et al., 2010): which rule preserves the visibility of a single catastrophic indicator without letting noise dominate, and on what grounds?

Challenge 2: Reliability of Declared Urgency

Eliciting the declared urgency U_d via pairwise multi-criteria methods introduces cognitive noise and ranking instability. These methods are sensitive to human inconsistency and rank reversal, meaning that declared urgency cannot be assumed as a deterministic ground truth (Afrasiabi et al., 2022; Triantaphyllou; Yanase, 2024; Liang et al., 2020). Operator judgments must therefore be treated as noisy, stochastic inputs (Stam et al., 1997): how can an elicitation protocol detect and reject inconsistent judgments before they contaminate the network state?

Challenge 3: Value-Added of the Adequacy Signal

Computing $\Delta U(t) = U_d - U_r(t)$ is necessary but not sufficient. The decisive scientific test is whether this adequacy signal improves the identification of operationally at-risk nodes beyond the raw variables already available to an operator (local KPI thresholds, unweighted urgency, or simple node degree) (Cohen et al., 2019; Bojko et al., 2021). If the score does not demonstrably outperform these simpler baselines against recorded outcomes, it serves merely as an auxiliary decision metric rather than a primary planning signal (Ahmetoglu, 2025; Turner et al., 2021).

Challenge 4: Composite Score Interpretability

Aggregating multiple criteria into a single composite score, whether the local urgency U_r across six KPI blocks or a ranking across a network of nodes, risks obscuring trade-offs and weakening decision interpretability. Composite-index literature warns that nominal weights do not always reflect real influence due to component correlations (Greco et al., 2018; Becker et al., 2017; Borgonovo et al., 2021): how can a coordinator still see *which* indicator is driving a node's urgency once everything is folded into one number?

Challenge 5: No Precedent for Urgency Adequacy

Digital twin systems routinely monitor real-time operational states, disruption responses, and order synchronization (Ding et al., 2023; Leung et al., 2022; J. Zhang et al., 2021; Y. Zhang et al., 2026; Zhou et al., 2021). However, to the best of our knowledge, no existing digital twin or decision support system simultaneously captures a client-declared urgency signal, computes a model-based operational urgency from live system state, and measures their mismatch as an active, network-propagated indicator. The absence of precedent cuts both ways: it signals an opportunity, but it also means there is no established architecture, calibration practice, or evaluation benchmark to build on (Zheng; Lu, et al., 2021; Niloofar et al., 2023; Kalaboukas et al., 2021).

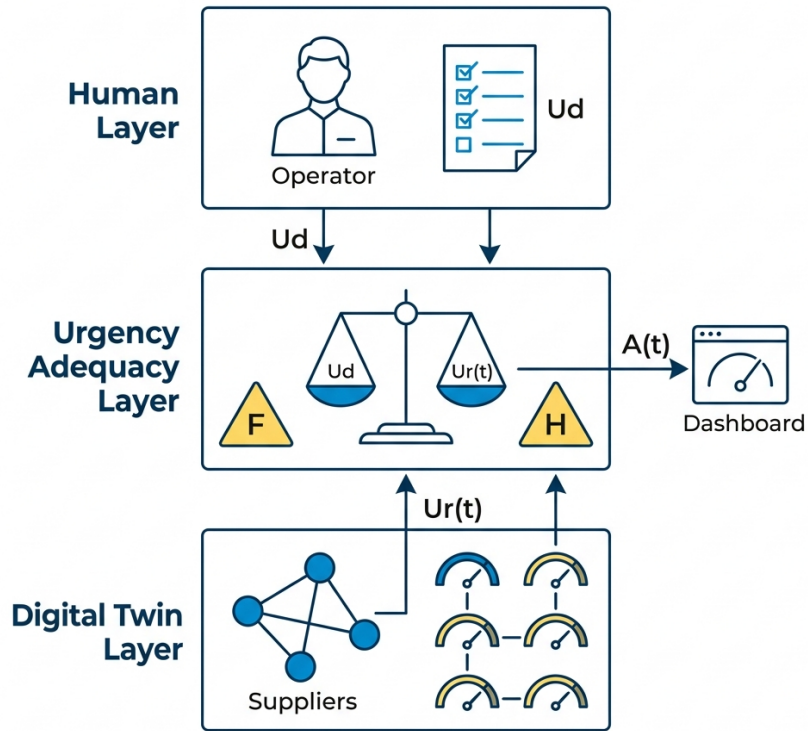


Figure 3: The three-layer urgency adequacy architecture investigated in this work (Challenge 5).

3 State of the Art

3.1 Introduction: Key Definitions

Urgency is defined here as the degree to which a supply chain task requires prompt action to avoid a material deterioration in operational performance. It differs from *importance*, which reflects strategic value, and from *priority*, which reflects the current execution order. In this work, urgency is treated as a multi-dimensional operational quantity, aggregated from heterogeneous KPI evidence rather than from a single deadline-driven signal (J. Chen et al., 2025; M. Zhu et al., 2018).

Real-time scheduling refers to the class of algorithms that assign execution order to tasks given temporal constraints (deadlines, inter-task dependencies), resource constraints (vehicle capacity, supplier availability), and optimality criteria (makespan, weighted tardiness) (Pinedo, 2012). This literature is drawn on here only as historical grounding for deadline-sensitive value functions; the present work does not deliver a scheduling algorithm.

Digital twin is a synchronized digital representation of a physical system that supports monitoring, simulation, and decision-making using real-time and historical data (Ivanov; Dolgui, 2020). In supply chains, digital twins are used to improve visibility, disruption assessment, and replanning under changing conditions (Ivanov; Dolgui, 2020; Ding et al., 2023; Hrušovský et al., 2021).

Multi-criteria decision-making (Multi-Criteria Decision-Making (MCDM)) covers methods that combine multiple, potentially conflicting performance dimensions into a decision (Saaty, 1980; Brans; Vincke, et al., 1986). In the present work, the main dimensions considered for node prioritisation are time pressure, capacity saturation, operational performance, failure risk, cost drift, and carbon overshoot.

Physical Internet (Physical Internet (PI)) is the model of open, mutualized logistics networks that route shipments through shared infrastructure using standardised modular containers (Montreuil, 2011). In such shared logistics environments, inaccurate urgency assessment may have effects beyond a single actor by influencing the allocation of mutualized capacity. This makes urgency evaluation particularly important in Physical Internet settings (Montreuil, 2011; Leung et al., 2022).

Serious game is a simulation environment designed to train decision-makers through realistic scenarios with immediate feedback and scoring (Sterman, 1989; Prensky, 2001). In this project, DISCO functions both as a serious game and as a digital twin: users interact with simulated supply chain decisions, while the underlying digital model evaluates operational consequences on a weekly cadence.

3.2 Literature Review

This review sets out to answer a single question: which existing methods can be assembled to measure, correct, and propagate a biased urgency signal, and what part of that assembly is missing from the literature? The relevant work is broad but fragmented across operations research, behavioural science, and digital twin engineering. Read individually, each stream solves one piece of the problem; read together, they leave a specific slot empty. No retrieved work combines, within a single supply-chain decision architecture, the five elements this thesis requires: (i) a multi-block operational urgency function, (ii) a structured client-side urgency input, (iii) an adequacy layer comparing stated and operational urgency, (iv) network propagation of that urgency across a multi-tier DAG, and (v) empirical validation of the adequacy signal against recorded outcomes (Sawik, 2013; Sawik, 2014; Sawik, 2016; Cohen et al., 2019; Hosseini; Ivanov, 2020).

The review is therefore organised by the role each stream plays in this argument, rather than by chronology or by method family. Two *foundational* streams establish why the problem exists: deadline-sensitive value and network risk propagation on one side, cognitive bias in urgency declaration on the other. Three *enabling* streams supply the computational toolkit: Bayesian belief revision, digital twin operations, and multi-criteria decision methods. One stream gathers the *closest antecedents*, whose precisely identified gap isolates what this work adds. Finally, *analogical* literature (medical triage, the Physical Internet, carbon accounting) provides vocabulary and context only, and is flagged as non-foundational wherever it is used.

Table 6 summarises the seven streams and their relationship to the thesis.

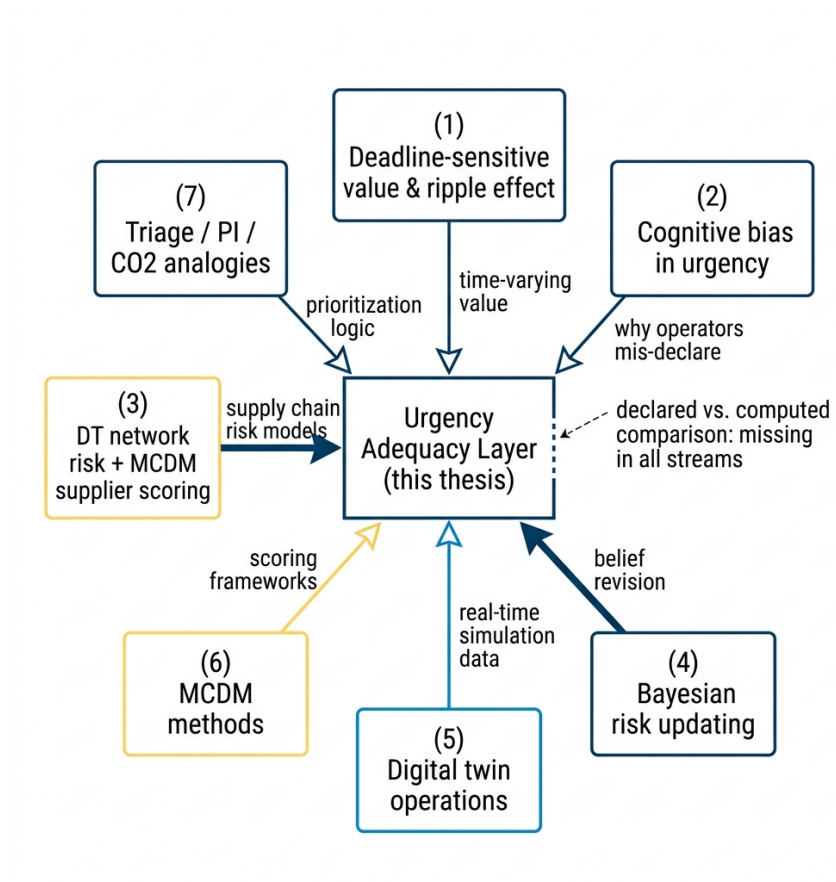


Figure 4: Map of the seven literature streams and the gap they leave open.

Table 6: Literature stream synthesis: contribution and gap relative to this thesis.

Stream	Role	Main contribution
Risk propagation and deadline-sensitive value	Foundation	Monotone time-value functions; ripple-effect propagation on networks
Cognitive bias in urgency	Foundation	Documents urgency mis-declaration patterns
Digital twins of network risk and MCDM supplier scoring	Closest antecedent	Multi-criteria supplier scoring and network disruption monitoring
Bayesian and probabilistic risk updating	Enabling method	Conjugate/probabilistic belief revision under new evidence
Digital twin operations	Enabling method	Real-time replanning and disruption response
MCDM methods	Enabling method	Criteria weighting and outranking
Medical triage / PI / CO ₂	Analogy / Context	Asymmetric urgency vocabulary; network context

3.2.1 Stream 1: Risk Propagation and Deadline-Sensitive Value

Role: Foundation.

Why: It demonstrates that operational value can be formulated as a time-dependent and network-dependent quantity, providing the primary motivation for treating urgency as a variable that is both time-varying and propagated across a multi-tier network rather than computed at a single node in isolation.

Within real-time systems, deadline-sensitive scheduling theory models task value as a function of completion or response time, an approach known as time-value or utility-accrual scheduling (K. Chen et al., 1996; J. Chen et al., 2025). Chen and Mühlethaler (K. Chen et al., 1996) show that task utility is typically constant up to a deadline and then non-increasing with delay; the utility-accrual survey by Chen et al. (J. Chen et al., 2025) confirms that deadline-sensitive literature models task value as a function of response time with bounded, structured degradation after critical thresholds. This establishes the historical precedent for treating urgency as time-varying, but it stops at a single task in isolation.

A second, complementary line of evidence comes from network risk-propagation

research. Ivanov and colleagues formalise the ripple effect, whereby a local disruption at one supplier cascades through downstream tiers and amplifies as it travels along the network (Ivanov; Dolgui, 2020; Ivanov; Dolgui, 2019; Li et al., 2020). Probabilistic risk-inference methods for supply chain systems (Zheng; L. Zhang, 2020a) further show that aggregating heterogeneous risk indicators into a single node-level score is itself an open modelling question, distinct from how that score should then be propagated.

Table 7 compares four candidate aggregation and growth models found across these two lines of literature.

Table 7: Comparative urgency aggregation and growth models and their properties.

Model	Formula	Key property	Use case
Linear (time)	$U_0 + \lambda t$	Constant pressure	Administrative tasks
Hyperbolic (time)	$K/(t_c - t)$, bounded via $\tanh$	Finite-time singularity, discounting	Synthetic scenario generation
Weighted average (blocks)	$\sum_m \omega_m u_m$	Fully compensatory: one weak block hidden by strong ones	Simple composite indices
Noisy-OR (blocks)	$1 - \prod_m (1 - u_m)^{\omega_m}$	Non-compensatory: one saturated block dominates	Multi-block operational risk

This thesis adopts the noisy-OR aggregation over the fully compensatory weighted average, on the grounds that a single catastrophic KPI block (for instance a complete capacity failure) should not be masked by acceptable scores on the remaining blocks, a property established in the probabilistic-inference literature on combining independent causal evidence (Pearl, 1988). The bounded hyperbolic discounting curve is retained separately, purely as an auxiliary tool for generating realistic synthetic urgency trajectories used in testing and calibration; a structurally similar acceleration appears in Log-Periodic Power Law (Log-Periodic Power Law (LPPL)) models of financial crash precursors (Johansen et al., 2001; Johansen et al., 2000), used here strictly as an analogical reference for that auxiliary generator, since empirical analyses by Feigenbaum (Feigenbaum, 2001) and Brée et al. (Brée et al., 2010) show that LPPL-family models are highly parameter-sensitive and unsuited to direct use as a production risk model.

Summary of Stream 1:

- **Established literature:** Deadline-sensitive scheduling supports time-dependent task valuation (J. Chen et al., 2025); ripple-effect research supports network propagation of local disruptions (Ivanov; Dolgui, 2020; Li et al., 2020).
- **Research gap:** Neither stream specifies how to aggregate heterogeneous, block-structured operational risk indicators at a single node before propagating them.
- **Working assumption:** A non-compensatory, noisy-OR aggregation of KPI blocks is proposed as the local urgency model, to be propagated with a leaky product rule across the network.
- **Systemic gap:** While these streams model network propagation and deadline sensitivity, neither addresses user-declared urgency or the mismatch between stated and operational urgency.

3.2.2 Stream 2: Cognitive Bias in Urgency Declaration

Role: Foundation.

Why: It establishes that client-declared urgency is systematically biased and cannot be taken at face value as a planning input.

Behavioural operations research suggests that human decision-makers systematically misrepresent urgency, a finding with direct implications for any system that relies on operator-declared urgency as an input.

Two convergent lines show this from different angles. Zhu, Yang and Hsee (M. Zhu et al., 2018) report the Mere-Urgency Effect in five controlled experiments: participants consistently over-prioritised imminent low-value tasks over distant high-value tasks, and the bias persisted even when participants were informed of its existence. Steel (Steel et al., 2006) provides a complementary formal model through Temporal Motivational Theory (Temporal Motivational Theory (TMT)):

$$V(t) = \frac{U \times E}{1 + \Gamma \times D(t)^k}, \quad (3)$$

where U is objective utility, E is the expectancy of success, $D(t)$ is the delay to deadline, and Γ , k are individual discounting parameters. As the delay to deadline shrinks toward zero, the motivational value $V(t)$ converges to the full product of utility and expectancy: immediacy becomes more salient as the deadline approaches. Near-term tasks can therefore attract disproportionate attention even when their broader objective value is lower (Steel et al., 2006), which helps explain why imminent tasks may be over-declared as urgent. The operational roles of the parameters in this formulation are structured in Table 8.

Table 8: Temporal Motivational Theory (TMT) parameters and domains.

Symbol	Concept	Operational Role
$V(t)$	Temporal motivational value	Dynamic task salience
U	Objective utility	Task strategic importance
E	Expectancy of success	Probability of task completion
$D(t)$	Delay to deadline	Remaining safe execution window
Γ	Impulsiveness parameter	Operator sensitivity to delay
k	Sensitivity scaling parameter	Non-linear scaling of delay discounting

These laboratory findings are supported by evidence from operational settings. Rusou, Amar and Ayal (Rusou et al., 2020) show that operators under workload preferentially complete small, easy tasks, the operational analogue of under-triage for large-scope supply chain activities. DiwasSingh et al. (DiwasSingh et al., 2020) confirm in a large field study that completing easy tasks first significantly hurts overall performance. Hospital admission control under occupancy uncertainty (Kim et al., 2020) produces over- and under-triage patterns analogous to those in logistics operator decisions under uncertain lead times. The magnitude of these biases in industrial supply chain contexts remains a subject of calibration. To mitigate these time-perception and workload biases, human-centered decision support systems (such as Rosenthal and Hiatt (Rosenthal et al., 2020)) use visual feedback techniques to show the mismatch between human plans and system temporal constraints, helping users calibrate their schedules.

This study treats declared urgency as a noisy signal and computes a model-based correction. The correction mechanism, and its empirical predictive validity, is detailed in Section 4.

Summary.

Known: Urgency over-declaration for near-deadline tasks is documented in controlled settings (M. Zhu et al., 2018) and in operational settings (DiwasSingh et al., 2020; Rusou et al., 2020).

Unknown: Bias magnitude in industrial supply chain contexts.

Assumed: Asymmetric penalty $\lambda_{\text{under}} > \lambda_{\text{over}}$ is the appropriate correction structure.

Open gap: Behavioural research diagnoses the problem. It does not prescribe a real-time operational correction.

3.2.3 Closest Antecedents: Digital Twins of Network Risk and MCDM Supplier Scoring

Role: Closest antecedents.

Why: These are the works most structurally similar to the thesis. Identifying their gap precisely isolates what this work adds.

The works most structurally similar to this thesis combine multi-criteria supplier scoring with disruption-aware network monitoring. Sawik (Sawik, 2013; Sawik, 2014; Sawik, 2016) develops mixed-integer programming models for resilient supplier order assignment integrating cost, service level, and disruption risk. Cohen, Grinshpoun and Stern (Cohen et al., 2019) and Afrasiabi et al. (Afrasiabi et al., 2022) extend this to MCDM supplier prioritisation under disruption, combining criteria weighting with allocation optimisation. Chakraborty et al. (Chakraborty et al., 2020) confirm that rule-based heuristics remain competitive against exact methods in dynamic industrial environments. These works are cited here for the closeness of their subject matter (multi-criteria, disruption-aware supply chain decisions), not because this thesis reuses their optimisation machinery, which addresses allocation and sequencing rather than urgency measurement.

Taken together, this stream models disruption-aware supplier evaluation with multi-criteria weighting. Two gaps distinguish it from the present work. First, none introduces a network-propagated urgency signal computed independently at every node and combined across tiers; risk is typically evaluated once, at the point of decision. Second, none compares a client-stated urgency against a model-derived urgency: the client-declared signal, where it exists, is used directly, without validation against live operational proxies.

3.2.4 Enabling Method: Bayesian and Probabilistic Risk Updating

Role: Enabling method.

Why: Revising a failure probability from a small number of new observations is a component of operational risk state. Bayesian conjugate updating is the borrowed tool.

Bayesian methods have emerged as a well-supported approach for revising risk estimates from new evidence in supply chain research (Pearl, 1988; Koller et al., 2009). Rather than a full Bayesian network with an explicit conditional-probability-table structure, this study draws on the narrower and more tractable case of Beta-Bernoulli conjugate updating: a failure probability p is treated as a Beta-distributed belief with an equivalent prior sample size n_0 , revised from a small number of newly observed equivalent trials (k, n) each time a calibrated operational event is

declared. This is the standard conjugate-prior construction for binomial-outcome belief revision (Pearl, 1988; Koller et al., 2009).

Three convergent findings from the retrieved Bayesian supply-chain literature support this choice of tool. Hosseini and Ivanov (Hosseini; Ivanov, 2020) provide the fullest survey of Bayesian methods applied to supply chain risk and resilience, finding that probabilistic belief-revision methods are often preferred to simple deterministic scoring when new, sparse evidence must update an existing risk estimate. Huang et al. (Huang et al., 2020) and Deng, Zhang and Wei (Deng et al., 2020) develop Bayesian models for disruption and delay probability estimation that serve as a methodological precedent for revising a KPI block's failure probability from a declared event. Lockamy (Lockamy et al., 2011) and Qazi et al. (Qazi et al., 2017) further support Bayesian revision as a practical tool for dynamic supply chain risk quantification under sparse, event-driven evidence, consistent with the small-sample setting of a weekly-cadence digital twin. Plain frequency counts, by contrast, are unstable when only a handful of events are declared per node per project, which motivates the conjugate-prior formulation adopted here over a raw frequentist estimate (Zheng; L. Zhang, 2020b).

The specific conjugate-update parameters used for each event family are developed in Section 4.

Summary.

Known: Bayesian conjugate updating is an established, low-data-regime tool for revising failure probabilities in logistics contexts.

Assumed: A shared, moderately informative prior (equivalent to roughly six months of weekly observations) is sufficient for initial calibration.

Open gap: Bayesian belief revision quantifies disruption exposure but does not, by itself, compare it against client-declared urgency.

3.2.5 Enabling Method: Digital Twin and Real-Time Decision Support

Role: Enabling method and primary gap source.

Why: DT systems are the architectural family closest to DISCO; this review establishes both what they already do and the specific gap this thesis addresses.

The digital twin concept was formalised by Grieves (Grieves, 2014) as a virtualised representation of a physical process. Tao et al. (Tao et al., 2018) extended the concept to manufacturing; Fuller et al. (Fuller et al., 2020) surveyed the enabling technologies behind modern deployments. In supply chain management, Ivanov et al. (Ivanov; Dolgui, et al., 2021) define the Supply Chain (SC) Digital Twin (DT) as integrating simulation (what-if analysis), optimisation (decision support), and

machine learning (pattern detection). Ivanov and Dolgui (Ivanov; Dolgui, 2020) address the use of digital SC twins for disruption risk and resilience management in the Industry 4.0 era. Wang, Wang and Liu (Wang et al., 2020) demonstrate DT-driven planning in a manufacturing context analogous to DISCO's satellite component network. More recently, in the context of Industry 5.0, these systems have evolved towards Human-in-the-Loop (HITL) Cognitive Digital Twins (CDTs) where semantic models and reasoning methods are used to capture human feedback and synchronize mental models with simulated physical operations (Niloofer et al., 2023).

Real-time freight planning work demonstrates that operational urgency proxies (slack, lateness risk, and shortage exposure) are present within these systems (Hrušovský et al., 2021; Perez et al., 2022). Model-predictive control approaches for vehicle routing (Karam et al., 2022) and digital-twin disruption response (Ding et al., 2023; Leung et al., 2022; X. Zhu et al., 2025) confirm the state of the art in live operational replanning.

Taken together, the digital-twin literature supports simulation, optimisation, and disruption-aware decision support. In the retrieved literature, however, no system surfaces a standard module that:

- captures a client-declared urgency signal as a structured input,
- computes a model-based operational urgency from the live system state, and
- measures their mismatch as an active, network-propagated indicator.

This three-layer architecture appears to be an open slot in digital SC operations (stated as an observation from the retrieved literature, not as a claim about the totality of published work). Whether filling this gap improves the identification of at-risk nodes is the central empirical hypothesis of this thesis (Challenge 3, Section 2.5).

3.2.6 Enabling Methods: Multi-Criteria Decision Methods

Role: Enabling methods.

Why: These multi-criteria decision-making (MCDM) tools provide the computational foundation for criteria weighting, outranking, and network prioritisation.

This research integrates and builds upon two distinct methodological domains within multi-criteria decision-making: pairwise comparison methods for eliciting weights, and outranking aggregation for turning those weights into a defensible ranking of nodes.

Pairwise Elicitation and Pairwise Comparison Methods. To construct a structured representation of operator-declared urgency (U_d), criteria weights must be systematically elicited. The Analytic Hierarchy Process (Analytic Hierarchy Process (AHP)) (Saaty, 1980) derives criteria weights as the principal eigenvector of a pairwise comparison matrix, validating logical consistency via the Consistency Ratio ($ConsistencyRatio(CR) \leq 0.10$). While widely applied in logistics, AHP suffers from rank reversal, where the introduction of a new alternative alters the relative ranking of existing ones, and is sensitive to cognitive noise during expert elicitation (Dyer, 1990; Triantaphyllou; Yanase, 2024).

To minimize operator cognitive load and improve consistency, the Best-Worst Method (Best-Worst Method (BWM)) (Rezaei, 2015) reduces the number of required pairwise comparisons from $n(n-1)/2$ to $2(n-1)$ by anchoring comparisons around chosen "best" and "worst" criteria. This is further extended by the Fuzzy Best-Worst Method (Fuzzy Best-Worst Method (F-BWM)) (S. Guo et al., 2017; Yucesan et al., 2019; Amiri et al., 2020), which uses Triangular Fuzzy Numbers (Triangular Fuzzy Numbers (TFNs)) to absorb human judgment uncertainty. In this study, consistent FBWM weight vectors (Liang et al., 2020) serve as an alternative elicitation gate to crisp AHP, and both feed the same downstream propagation and adequacy machinery.

Outranking Aggregation, Network Criticality, and Composite Indicator Critique. For prioritising which nodes require attention, direct weighted averages risk full compensatory aggregation, where an excellent score on one criterion (e.g., cost) completely masks a critical failure on another (e.g., capacity). To prevent this, outranking methods such as PROMETHEE II (Brans; Vincke, et al., 1986; Brans; Mareschal, 2005) establish pairwise preference flows (ϕ^+ and ϕ^-), allowing a non-compensatory ordering of alternatives. A survey of PROMETHEE applications across industries (Behzadian et al., 2010) confirms its suitability for supplier and node evaluation. In this work, PROMETHEE II ranks active network nodes directly on the same weighted KPI blocks that feed the local urgency aggregation, giving a network-wide outranking view that complements the per-node adequacy score. Systematic what-if shock simulation, in which a node's local risk is forced to saturation and the resulting propagated delta is measured across the network, provides a second, independent criticality signal that is combined with the outranking flow to identify structural bottlenecks.

However, constructing any composite indicator from weighted criteria introduces methodological challenges. Composite-index literature warns that nominal weights assigned to criteria often diverge from their effective importance due to correlation, normalization, and aggregation dynamics (Greco et al., 2018; Becker et al.,

2017; Borgonovo et al., 2021), and the choice of normalization method is itself non-neutral: it can lead to rank reversal (Triantaphyllou, 2000). To keep the aggregation weights interpretable and defensible, this study exposes per-block contributions on the operator explainability page rather than only the aggregate score, while validating the risk parameters under uncertainty (Sawik, 2020).

3.2.7 Analogical and Contextual Literature

Role: Analogies and context.

Why: These bodies of literature provide conceptual analogies (medical triage), network contexts (Physical Internet), or regulatory frameworks (carbon accounting) that support the design of the adequacy variables.

While the core scoring and network-propagation methods are mathematically derived, their operational structure draws grounding and context from three non-foundational literature streams.

Medical Triage as a Conceptual Analogy. Medical triage provides a rich conceptual analogy for prioritizing tasks under severe capacity constraints and cognitive pressure. Standard protocols such as the Manchester Triage System (Manchester Triage System (MTS)) (Mackway-Jones et al., 2013) categorize patients into five urgency levels using flowcharts, whereas the Emergency Severity Index (ESI) integrates predicted resource consumption. Comparative clinical studies (Kuligowski et al., 2025) show significant classification discrepancies (weighted $\kappa = 0.51$) between these systems, highlighting that even trained professionals struggle with consistent urgency assessment.

In a supply chain context, this triage process maps directly to node prioritisation. The design of the cognitive adequacy score $A(t)$ is motivated by the clinical principle of asymmetric risk: under-triage (under-declaring urgency, leading to untreated critical conditions) is far more dangerous than over-triage (over-declaring urgency, leading to resource congestion) (Kim et al., 2020). Consequently, the adequacy score enforces an asymmetric penalty ($\lambda_{\text{under}} > \lambda_{\text{over}}$) to protect the physical supply chain from hidden stockout ruptures. Table 9 maps these triage concepts to their logistics equivalents.

Table 9: Heuristic mapping of medical triage concepts to DISCO logistics prioritisation.

Medical concept	DISCO logistics equivalent
Patient priority P1 (immediate)	Node with $A(t) < 20$ and high network criticality rank
Under-triage (P3 classified as P1)	Hidden risk $H = [U_r - U_d]_+$: silent bottleneck builds unnoticed
ESI resource consumption estimate	DISCO criticality index (propagated impact on final-customer nodes)
Resource-blind triage (MTS)	Local urgency threshold ignoring network position
Resource-aware triage (ESI)	Criticality-weighted prioritisation of coordinator attention

Physical Internet and Supply Chain Sychromodality. The Physical Internet (PI) (Montreuil, 2011) conceptualizes open, mutualized logistics networks where routing, storage, and transport assets are shared. In such hyperconnected systems, urgency ceases to be a localized concern and becomes a network-level Quality of Service variable. Dynamic routing and order synchronization in sychromodal networks require real-time, responsive priority signals compatible with rolling planning windows (W. Guo et al., 2020). Disruptions also propagate rapidly through hyperconnected networks, creating cascading ripple effects (Li et al., 2020) that validate the need for a synchronized, digital-twin-driven propagation model (Ivanov; Dolgui, 2020).

Regulatory Carbon Accounting. The integration of an environmental criterion is motivated by modern corporate sustainability requirements, such as the Corporate Sustainability Reporting Directive (CSRD Scope 3) and ALTEN’s environmental guidelines (Cordova, 2024). Standardized frameworks like the GLEC Framework v3.2 (Smart Freight Centre, 2023) and Vehicle Energy Consumption calculation TOol (VECTO) (European Commission, 2017) define emissions calculation baselines for heavy duty vehicles. ADEME (ADEME, 2014) reports that emissions factors are non-linearly related to vehicle speed and cargo load, supporting the treatment of carbon overshoot as its own urgency block rather than folding it into a generic cost term.

3.2.8 Validation Methodology: Serious Games

Role: Validation rationale.

Why: It justifies the weekly-cadence serious game used as the validation environment for the whole framework.

The Beer Game (Sternan, 1989) is the foundational supply chain serious game: Sternan demonstrates that players systematically under-react to upstream demand signals, generating the bullwhip effect. This decision bias is analogous to the under-triage pathology in logistics, where operators consistently under-respond to signals of building urgency. The finding that game-based observation captures systematic decision biases supports the use of digital twin serious games as an empirical instrument. Prensky (Prensky, 2001) establishes the theoretical basis for digital game-based learning, and Deterding et al. (Deterding et al., 2011) formalise gamification design principles.

Mohaddesi et al. (Mohaddesi et al., 2020) introduce the Gamette methodology, a playful, bite-sized approach to capturing decision-making patterns to build behavioural models, directly applicable to a weekly-review protocol in which operators repeatedly submit urgency assessments over the course of a project. Bellotti et al. (Bellotti et al., 2013) provide an evaluation framework for serious game learning outcomes, which supports the design of the calibration methodology detailed in Section 4.

3.3 Summary: Limitations of Existing Solutions

A review of the literature indicates that classical operational monitoring typically evaluates a single criterion (a deadline, a cost, or a risk score) at a time, failing to capture the multi-block nature of supply chain KPIs or the systemic impact of operator decision biases. While pairwise comparison methods (AHP, BWM, and Fuzzy BWM) are widely applied for criteria weighting, they function as static elicitation tools and are rarely compared against a computed operational counterpart. Outranking methods like PROMETHEE II prevent compensatory aggregation but are not usually combined with a network-propagated urgency signal.

While contemporary supply chain digital twins and Bayesian risk models offer strong capabilities in real-time disruption monitoring and belief revision, they typically lack a mechanism to compare human operator inputs with the physical state of the digital twin. This review identifies a key gap: the absence of an integrated architecture that compares a declared client-side urgency signal against a computed, network-propagated operational urgency to evaluate and correct cognitive mismatch, and that validates this correction empirically against recorded

operational outcomes. Whether this adequacy-driven prioritisation improves the identification of at-risk nodes compared to standard KPI baselines is the central open question of this R&D operation.

3.4 Research Axis and Proposed Adequacy-Propagation Structure

To investigate this gap, this study proposes an architecture centered on a cognitive-physical adequacy check combined with network-wide risk propagation. The main objective is to design and evaluate a two-layer control loop:

1. **Cognitive adequacy check:** This layer captures the operator's declared urgency U_d (elicited through consistency-gated pairwise comparisons) and compares it with a physically-grounded real urgency $U_r(t)$ computed from live digital twin state. The mismatch is processed using an asymmetric penalty to debias operator judgments.
2. **Network propagation and criticality-driven prioritisation:** Local urgency is propagated across the supply chain DAG (declared urgency descending toward suppliers, real urgency ascending toward the final customer), and the resulting network state feeds both a PROMETHEE II outranking ranking and a systematic shock-simulation criticality index, jointly identifying which nodes warrant prioritised operator attention.

This two-layer architecture is exercised and empirically validated within a persistent, multi-tenant digital twin operated as a weekly-cadence serious game, closing the loop between the adequacy signal and observed operational outcomes (Section 4).

4 Conducted Operations

4.1 Methodology: The Agile Test-and-Learn Program

This R&D operation is structured as an agile Test-and-Learn program designed to investigate cognitive biases in supply chain urgency assessment and formulate a digital-twin-based correction loop. Rather than following a rigid sequential path, the program operates through continuous iterations of literature analysis, mathematical specification, software implementation, and scenario-based simulation.

The R&D pipeline is organized into four core operational areas:

- **Literature Review and Gap Analysis:** Establishing the psychological and mathematical foundations of multi-block urgency aggregation and cognitive bias (detailed in Section 3).
- **Mathematical Framework Specification:** Deriving and formalizing the multi-block real urgency function, its DAG propagation, and the asymmetric cognitive adequacy score from first principles to ensure dimensional and probabilistic consistency.
- **Pairwise comparison elicitation:** Formulating crisp Analytic Hierarchy Process (AHP) and Fuzzy Best-Worst Method (F-BWM) eigenvectors to build a reliable representation of declared operator urgency.
- **Implementation, Simulation and Validation:** Building a modular, tested Python library (65 modules, 1425 automated tests) exercised through a multi-page Dash web application, running controlled shock and scenario simulations on a persistent digital twin, and confronting the hidden-risk signal with recorded outcomes through a dedicated calibration service and a serious-game campaign (Section 4.8).

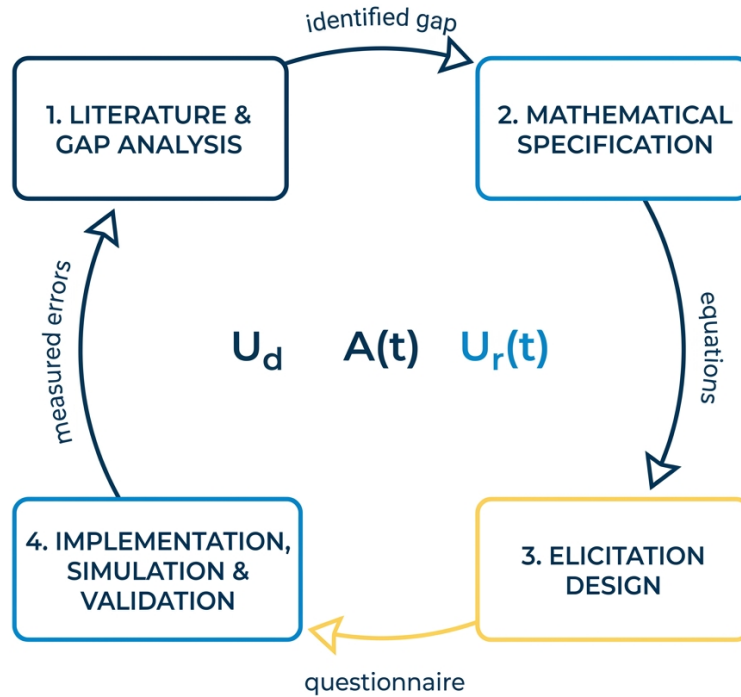


Figure 5: The agile Test-and-Learn iteration loop of the R&D operation.

4.2 System Architecture

Before the mathematical content, the delivered system’s architecture deserves a short description of its own: several design guarantees relied upon later (auditability, multi-tenant isolation, deterministic testing) are architectural, not mathematical.

Single facade. All operations go through one orchestrating facade (`services/orchestrator.py`), which coordinates the mathematical core, the graph repository, and the persistence layers behind a single thread-safe entry point. The web interface, the command-line entry point, and the test suite all drive the system exclusively through this facade; there is no second write path.

Multi-tenant persistence. Data is split between one global registry database (projects, nodes, arcs, milestones, scenarios, registry-level audit) and one SQLite database per node holding its private operational history (KPI snapshots, AHP assessments, weekly reviews, urgency history, declared events, and a per-node audit log). This isolation prevents write contention between operators reviewing different nodes and enforces the boundary that one supplier's raw history is not readable through another supplier's records.

Gatekeeper writes. Every business write crosses a single mutation service that whitelists the touched fields, validates ranges, diffs against the current state (a no-op write leaves no trace), records the change in the audit log with its author and source, and synchronises the in-memory graph. Audit tables are strictly immutable, including from the administration interface: the "no silent data loss" guarantee is enforced here, not by convention.

Graph and propagation. The network lives in an in-memory graph repository (networkx-backed; a Neo4j adapter is scaffolded but out of scope). Propagation is incremental: only the upstream/downstream cones of nodes whose local urgency actually changed are recomputed, which keeps the twin responsive at the 1000+-node scale validated by dedicated performance benchmarks.

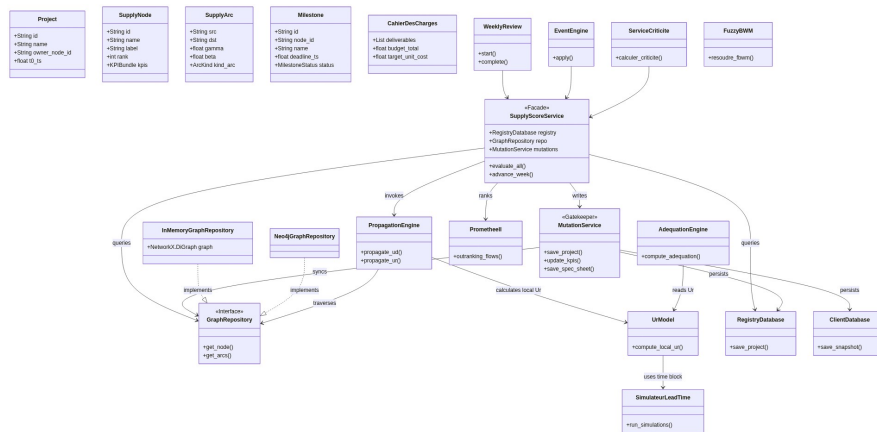


Figure 6: Doxygen-generated UML view of the SupplyScore architecture: single facade, gated writes, multi-tenant persistence.

Decoupled clocks. Each project carries its own clock: a system clock for live use, a game clock advancing week by week for serious-game sessions, and a fixed clock for deterministic tests. The weekly cycle, milestone pressure, and event timestamps all read the project clock, never the wall clock directly, which is

what makes a compressed serious-game campaign (Section 4.8) possible without touching the model.

4.3 Mathematical Framework and Core Formulations

4.3.1 Core Model: Multi-Block Real Urgency and Network Propagation

Local Real Urgency ($U_{r,\text{local}}$). Real urgency is not read from a single deadline-proximity signal; it is aggregated at each node from six independent KPI blocks, $\{\text{time, cap, perf, risk, cost, co2}\}$, each producing a partial urgency $u_m \in [0, 1]$ or `None` when the underlying KPI data is unavailable. The aggregation is a weighted probabilistic OR, so that a single saturated block is sufficient to drive the local urgency toward its maximum, regardless of how well the other blocks score:

$$U_{r,\text{local}} = 1 - \prod_m (1 - u_m)^{\omega_m}, \quad (4)$$

where $\omega_m \geq 0$ is the aggregation weight of block m , and the product runs over blocks with available data. Table 10 gives, for each block, its schematic formula and what it measures; the notation $[x]_+ = \max(x, 0)$ denotes the positive part, and the per-term weights (α_i, β_i) are renormalised over the terms whose data is actually available.

Table 10: The six real urgency blocks: formula and explanation.

Block	Formula	Explanation
u_{time}	$P(L > d^* - t) + \kappa_{\text{retard}}[r]_+ - \kappa_{\text{avance}}[-r]_+$	Probability that the lead time L , modelled as $\text{Normal}(\mu, \sigma)$, exceeds the time left before the next active deadline d^* ; forced to 1 once the deadline is passed. The planning term r compares theoretical progress with reported progress: a milestone behind plan raises the block, one ahead of plan lowers it.
u_{cap}	$\alpha_1 \left(1 - \frac{V_{\text{rem}}}{V_{\text{max}}}\right) + \alpha_2 \left(1 - \frac{W_{\text{rem}}}{W_{\text{max}}}\right) + \alpha_3 \left[\frac{D - \Phi}{D}\right]_+$	Saturation of the node's capacity: how little storage volume and weight remain, plus the fraction of demand D not covered by the current flow rate Φ .
u_{perf}	$1 - \text{OEE}$	Complement of Overall Equipment Effectiveness (availability $\times$ performance $\times$ quality): the lower the productive health of the critical resource, the higher the urgency.
u_{risk}	$1 - \exp(-p_{\text{fail}} \times t_{\text{recov}} \times \text{sev}/T_{\text{ref}})$	Expected unavailability: failure probability times recovery time times impact severity, normalised by a reference horizon T_{ref} , then combined by probabilistic OR with the environmental and political exposure terms.
u_{cost}	$\beta_1 \frac{c_{\text{op}} - c_{\text{nom}}}{c_{\text{nom}}} + \beta_2 [\text{tariff} - 1]_+ + \beta_3 \frac{c_{\text{stock}}}{c_{\text{ref}}}$	Financial stress: drift of operating cost above its nominal value, tariff multiplier above neutral, and storage cost relative to a reference cost.
u_{co2}	$\left[\frac{e - e_{\text{target}}}{e_{\text{max}} - e_{\text{target}}}\right]_+$	Normalised overshoot of measured emissions e above the target, saturating at the maximum threshold.

Two lifecycle status rules are applied before aggregation: a node marked **DONE** has its effective local U_r forced to 0.0 (a completed task carries no further physical risk), while a node marked **ABANDONED** has it forced to 1.0 (complete failure). These rules are applied identically whether the local urgency comes from the analytic blocks above or from the Monte Carlo estimate of the time block described below.

Why These Six Blocks: A Rupture-Risk-Driven Selection. The six blocks were not chosen as an arbitrary KPI shortlist. The selection started from the question

that defines urgency in a supply chain: *what mechanisms make the risk of a delivery rupture rise?* A rupture, that is the inability to deliver at the agreed time, place, quantity, and quality, is the terminal event the urgency signal must anticipate (Heckmann et al., 2015). Working backwards from that event, and consistently with indicator-based supply risk monitoring (Mokhtar et al., 2019), four *direct causal mechanisms* and two *arbitration variables* were identified:

- **Time erosion** (u_{time}): the remaining slack between deadline and realistic lead time. Time is the only non-compressible resource: whatever the state of the other indicators, a task whose slack reaches zero can no longer be saved, which makes deadline proximity the variable most directly tied to operational urgency (Cohen et al., 2019; Bojko et al., 2021; Sun et al., 2020; Sawik, 2013).
- **System saturation** (u_{cap}): queues, transport capacity, workshop load, and bottlenecks. A saturated system introduces hidden waiting delays on top of the nominal lead time, so capacity stress raises non-delivery risk even when the announced lead time looks safe (Tomlin, 2006; Hosseini; Tajik, et al., 2019; Taghavi et al., 2024).
- **Critical-resource performance** (u_{perf}): drift in the performance of the resource that produces the part. OEE is used as a composite proxy (availability $\times$ performance $\times$ quality) rather than a single machine metric, because precise performance data is rarely accessible on the supplier side (Mokhtar et al., 2019; Hlioui et al., 2017).
- **Adverse-event exposure** (u_{risk}): the expectation of unavailability caused by a disruption, computed as failure probability $\times$ recovery time $\times$ impact severity, composed with environmental and political exposure. The default incident probability (0.02) is revised weekly from declared events through the Bayesian operator of Section 4.6 over a $n_0 = 26$ iteration window; this block builds on the DIN supplier risk scorer (Kabouche, 2025) and on probabilistic ripple-effect models (Hosseini; Ivanov; Dolgui, 2020; Liu et al., 2021; Hosseini; Ivanov, 2020).
- **Financial flexibility** (u_{cost}): expediting surcharges, tariff rises, storage costs. Cost drift is not a rupture cause in itself but a stress indicator of the chain (paying premium express transport to avoid a physical rupture is a measurable symptom that the system is straining), and it bounds which corrective actions remain economically viable (Shen et al., 2019; Tomlin, 2006).
- **Carbon overshoot** (u_{co2}): emissions above target. Like cost, it is an arbitration criterion rather than a rupture mechanism: it only becomes decisive when the regulatory or ESG constraint genuinely restricts the set of admissible responses, typically the ship-vs-air trade-off (Hoen et al., 2014; Smart Freight Centre, 2023; International Organization for Standardization, 2023).

Each candidate indicator was screened against four filter questions, summarised in Table 11: only indicators passing all four were retained, which yields a set of four direct causes plus two arbitration variables and explains why the list is six blocks rather than four or eight.

Table 11: Selection filter applied to each candidate KPI block.

Criterion	Question asked of each candidate indicator
Causal link	Does the indicator have a plausible mechanism linking it to delivery rupture?
Observability	Can it be measured or credibly estimated from client, supplier, or planning data?
Actionability	Can an operator act on it, or use it to decide?
Non-redundancy	Does it add information not already carried by a retained indicator?

At this stage the selection remains a *conceptual* construction: it must still be confirmed by the real availability of the underlying data and by sensitivity tests verifying that each block contributes useful information to the aggregate (Section 5.1).

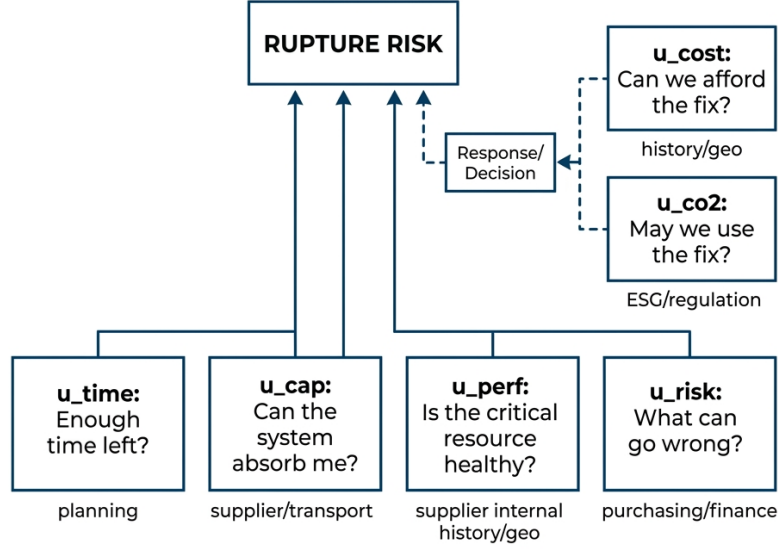


Figure 7: Causal structure behind the six-block selection: four direct rupture mechanisms and two arbitration variables.

Network Propagation. Declared urgency U_d propagates *downstream*, from the final customer (rank 0) toward raw-material suppliers, while real urgency U_r propagates *upstream*, from suppliers toward the final customer, using a leaky product rule that keeps both signals bounded in $[0, 1]$ and monotonically increasing as more neighbours become urgent:

$$U_{d,i} = 1 - (1 - U_{d,\text{local},i}) \prod_{k \in \text{Succ}(i)} (1 - \gamma_{ik} U_{d,k}), \quad (5)$$

$$U_{r,i} = 1 - (1 - U_{r,\text{local},i}) \prod_{j \in \text{Pred}(i)} (1 - \beta_{ji} U_{r,j}), \quad (6)$$

where $\gamma_{ik} \in [0, 1]$ is the downstream attenuation coefficient on arc (i, k) and $\beta_{ji} \in [0, 1]$ is the upstream amplification coefficient on arc (j, i) . Backup arcs are

purely documentary and inert with respect to both propagations. Table 12 details the propagation variables.

Table 12: Component definitions for the network propagation equations.

Component	Type	Description
$U_{d,i}, U_{r,i}$	Dependent variables	Propagated declared/real urgency at node i
$U_{d,local,i}, U_{r,local,i}$	Input variables	Local urgency at node i after status-rule adjustment
γ_{ik}	System parameter	Downstream (demand) attenuation coefficient on arc $i \rightarrow k$
β_{ji}	System parameter	Upstream (risk) amplification coefficient on arc $j \rightarrow i$
$Succ(i), Pred(i)$	Structural sets	Nominal (non-backup) successors and predecessors of node i

Propagation is recomputed incrementally: only the nodes downstream or upstream of a change (a submitted questionnaire, a revised KPI, a structural edit) are recalculated, rather than the full graph, which keeps the digital twin responsive as the network grows.

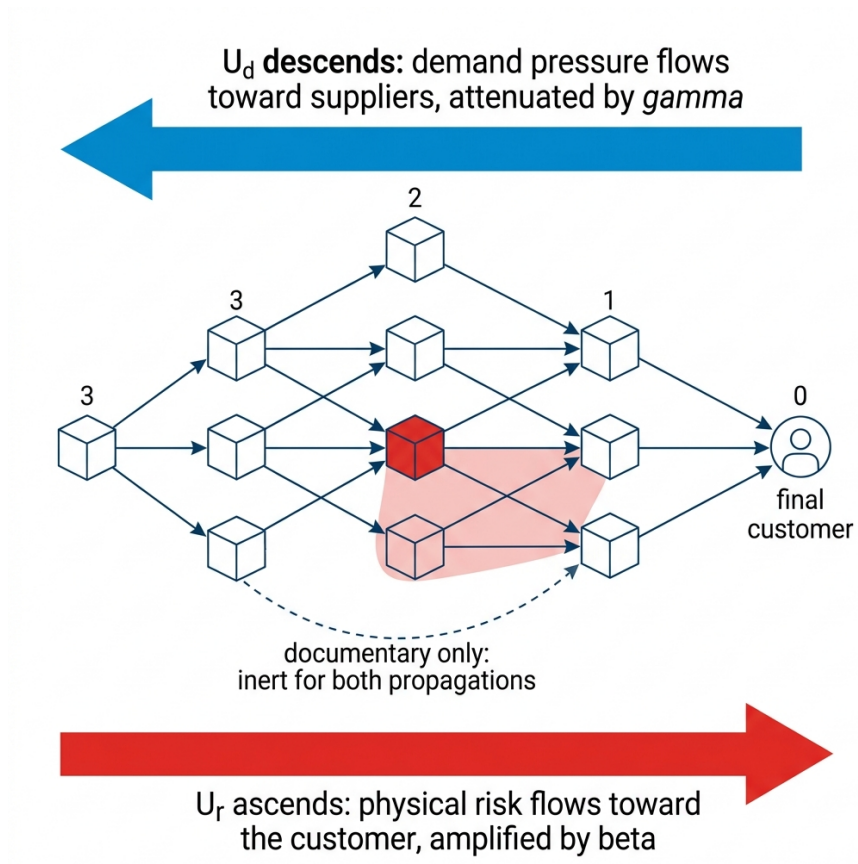


Figure 8: Bidirectional urgency propagation on the supply chain DAG.

4.3.2 Core Model: Asymmetric Cognitive Adequacy (Prospect Theory)

The mismatch between declared and real urgency is split into two named pathologies, false urgency F (over-declaration, panic) and hidden risk H (under-declaration, silent danger):

$$F = \max(U_d - U_r, 0), \quad H = \max(U_r - U_d, 0). \quad (7)$$

Drawing on Kahneman and Tversky's Prospect Theory (Kahneman; Tversky, 1979), the two pathologies are penalised asymmetrically and with diminishing marginal sensitivity to large errors:

$$\text{penalty} = \lambda_{\text{under}} H^\alpha + \lambda_{\text{over}} F^\alpha, \quad (8)$$

and the penalty is rescaled into an adequacy score $A \in [0, 100]$ through an exponential mapping bounded by the penalty of a maximal one-sided error:

$$A = 100 \times \frac{\exp(-\text{penalty}) - \exp(-\text{penalty}_{\max})}{1 - \exp(-\text{penalty}_{\max})}, \quad \text{penalty}_{\max} = \max(\lambda_{\text{under}}, \lambda_{\text{over}}). \quad (9)$$

Table 13 lists the parameters, with their governing constraint $\lambda_{\text{under}} > \lambda_{\text{over}}$: a coordinator who panics wastes premium transport capacity, but a coordinator who under-reports risk misses the window to prevent a line stoppage, so under-declaration is priced more expensively.

Table 13: Component and parameter definitions for the asymmetric adequacy equations.

Component	Type	Description	Default
F	Dependent variable	False urgency (over-declaration)	n/a
H	Dependent variable	Hidden risk (under-declaration)	n/a
λ_{under}	System parameter	Penalty weight on hidden risk	2.25
λ_{over}	System parameter	Penalty weight on false urgency	1.0
α	System parameter	Psychophysical curvature (diminishing sensitivity)	0.88
A	Dependent variable	Adequacy score, 100 = perfect alignment	[0, 100]

The default $\lambda_{\text{under}}/\lambda_{\text{over}} = 2.25$ ratio reprises the loss-aversion coefficient measured by Kahneman and Tversky (Kahneman; Tversky, 1979; Kahneman, 2011), and $\alpha = 0.88$ reprises their measured curvature of the value function. These defaults carry the status of literature-derived starting points, not measured properties of supply chain coordinators: they have not yet been validated empirically in this context, and their recalibration is one purpose of the validation campaign of Section 4.8. Both the aggregation weights ω_m and the penalty parameters $(\lambda_{\text{under}}, \lambda_{\text{over}}, \alpha)$ are configurable per project rather than hard-coded, so that this recalibration requires no code change.

4.3.3 Monte Carlo Estimate of the Time Block

The analytic u_{time} block assumes a Normal lead time at a single node. On a multi-tier network this assumption breaks: a node can only start when *all* its predecessors have delivered, and the maximum of several skewed lead times has no convenient closed form. The module `mc/lead_time.py` therefore replaces the analytic estimate with a stochastic one when enabled.

The simulation works in three steps. First, each node draws its lead time L_i from a configurable distribution (lognormal, normal clipped to zero, or triangular). Second, completion times propagate along the DAG in topological order: a node starts when its last predecessor finishes ($S_i = \max_{j \in \text{Pred}(i)} C_j$) and completes after its own lead time ($C_i = S_i + L_i$). Third, the delay probability of each node is estimated as the fraction of draws in which it misses its deadline, $u_{\text{time}} = P(C_i > d_i)$, over $N = 10\,000$ draws by default. A memory guardrail caps the simulation at $N \times n \leq 5 \times 10^7$ across n nodes.

The estimate feeds directly into Equation 4 through a dedicated override point, so the aggregation, status rules, and propagation machinery are reused unchanged.

4.3.4 Network Criticality Index

The criticality service (`services/criticite.py`) answers one question systematically: which node, if it failed completely, would cause the most damage to final-customer delivery? For every active node in turn, the service forces the node's local real urgency to 1.0, re-runs the ascending propagation, and measures two deltas: the rise of U_r at the final-customer nodes ($\Delta U_{r,\text{final}}$) and the largest rise anywhere in the network ($\Delta U_{r,\text{max}}$). Nodes are ranked by $\Delta U_{r,\text{final}}$ first, then $\Delta U_{r,\text{max}}$, then name.

A small example makes the mechanics concrete. Take a three-node chain, raw-material supplier $\rightarrow$ sub-assembly $\rightarrow$ final assembly, with upstream coefficients $\beta = 0.9$ then 0.7, and a baseline where the sub-assembly node carries $U_r = 0.40$ and the final node $U_r = 0.28$. Forcing the sub-assembly node to $U_r = 1.0$ propagates to the final node as $1 - (1 - 0.7 \times 1.0) = 0.70$, so its shock deltas are $\Delta U_{r,\text{final}} = 0.70 - 0.28 = 0.42$ and $\Delta U_{r,\text{max}} = 1.0 - 0.40 = 0.60$ (Figure 10).

One behaviour of this index matters for interpretation: a node already saturated at $U_r \geq 1.0$ yields a delta of zero and sinks to the bottom of the ranking, since its risk is already realised in the baseline. In a network-wide crisis, where many nodes saturate at once, the index therefore degenerates; this documented behaviour drives the exclusion rule used by hypothesis H3 in the validation campaign (Section 4.8).

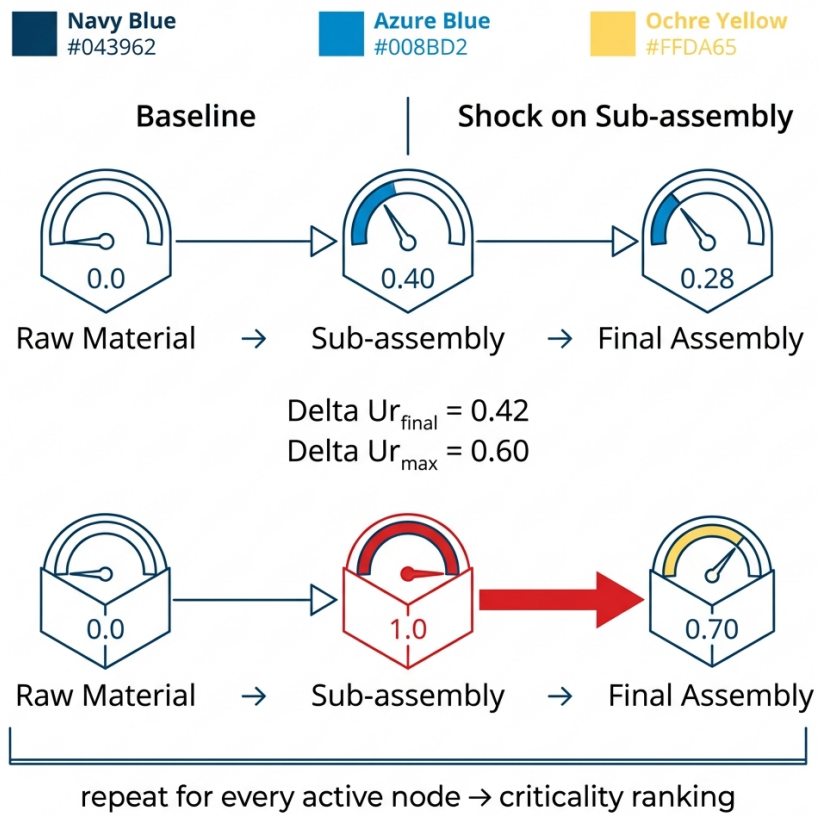


Figure 9: One iteration of the systematic shock simulation behind the criticality index.

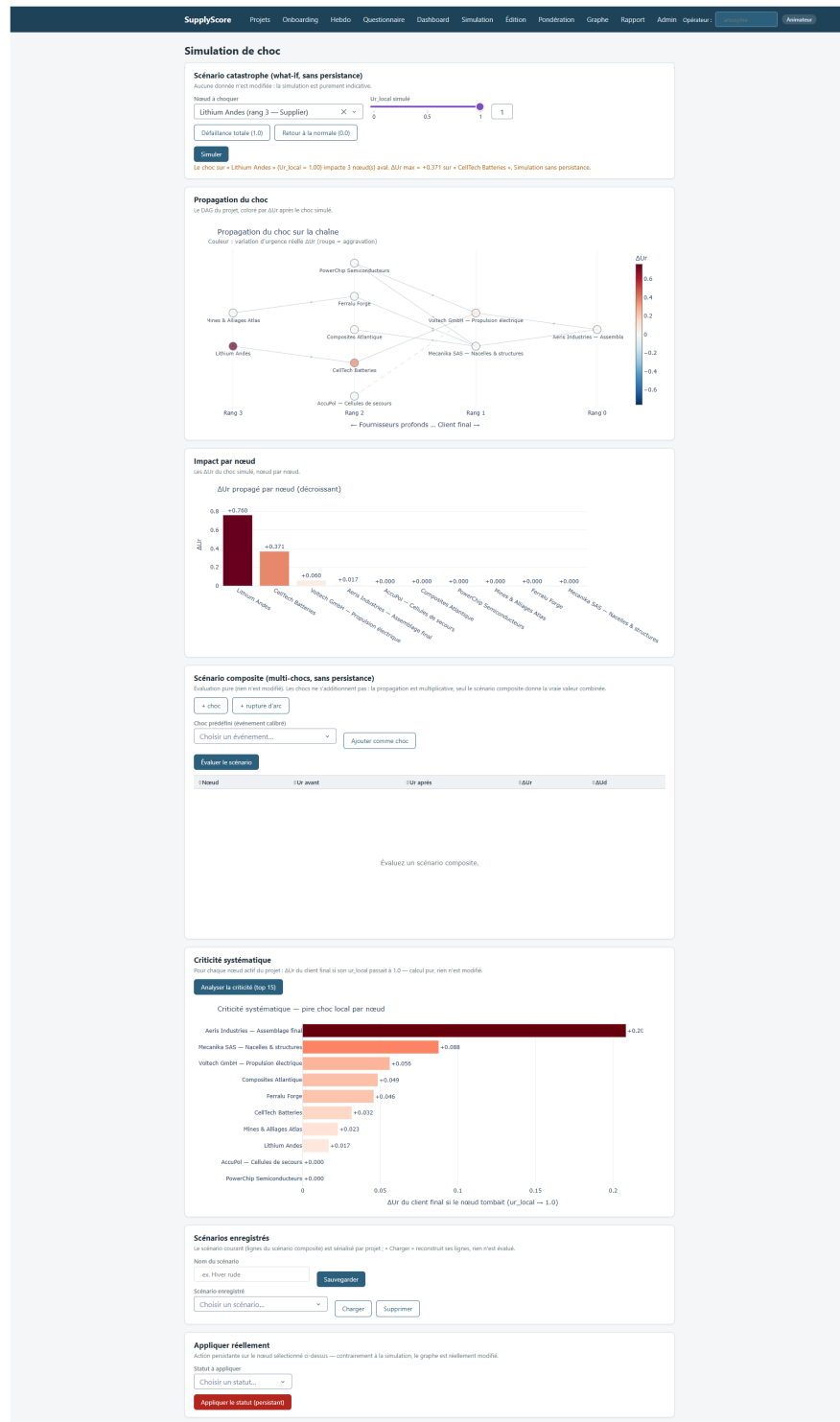


Figure 10: Network criticality ranking after a systematic shock simulation.

4.4 Expert Criteria Weighting: The AHP Elicitation Model

The Python module `core/ahp.py` implements the Saaty eigenvector method for a 4-criterion questionnaire administered through the weekly review page. The four criteria are:

- C1.** *Operational impact*: “If this task fails, what is the downstream consequence?”
- C2.** *Time window*: “How long before the situation becomes irredeemable?”
- C3.** *Downstream dependencies*: “How many tasks are blocked waiting for this one?”
- C4.** *Recoverability*: “Can we catch up if this task is late?” (inverted, where lower is worse)

How the Four Criteria Were Chosen. The criteria set follows from two constraints, one cognitive and one structural. Cognitively, Saaty’s method degrades as the number of pairwise-compared criteria grows: consistency becomes hard to maintain beyond roughly seven items, and each additional criterion adds $n - 1$ comparisons to the operator’s weekly workload (Saaty, 1990; Triantaphyllou; Yanase, 2024). Keeping $n = 4$ limits the questionnaire to six comparisons, short enough for a weekly cadence. Structurally, the four questions mirror the way declared urgency is consumed downstream: C1 (impact) and C3 (dependencies) capture the *severity* dimension that propagation amplifies across the network, C2 (time window) captures the *slack* dimension confronted with u_{time} , and C4 (recoverability) captures the *resilience* dimension confronted with the recovery term of u_{risk} . Each declared criterion therefore has an operational counterpart in the computed signal, which is what makes the later U_d vs. U_r comparison meaningful rather than a comparison of unrelated quantities. Larger criteria sets (cost, sustainability) are deliberately routed to the F-BWM weighting of the KPI blocks instead (Section 4.5), where the elicitation burden is lower.

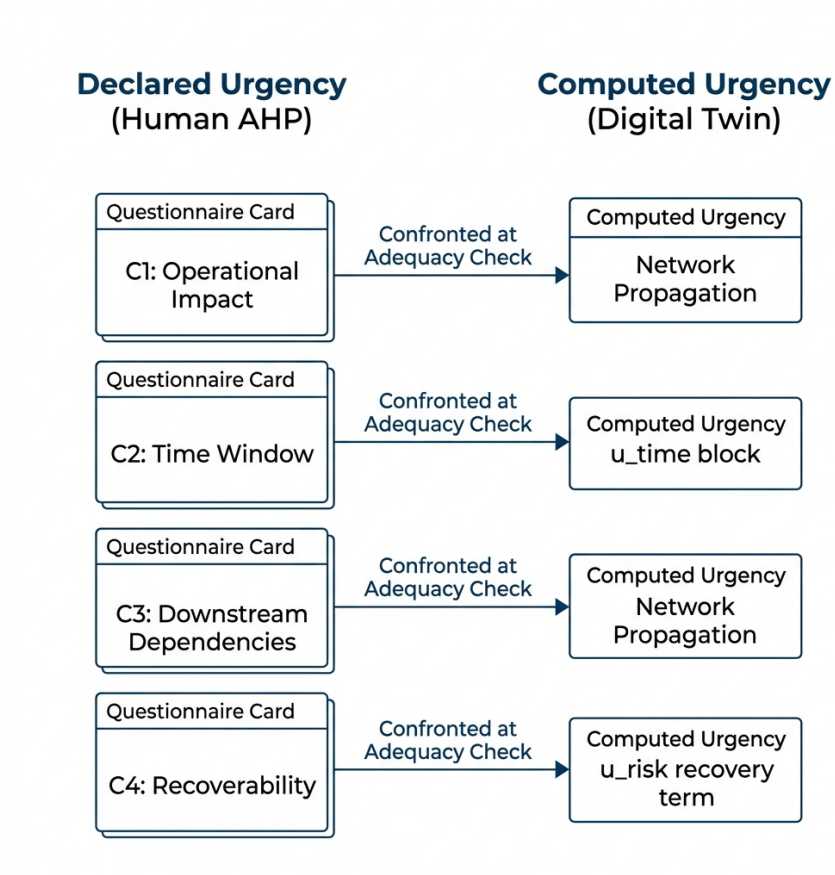


Figure 11: Correspondence between the four AHP criteria and their computed counterparts.

The consistency ratio is computed as:

$$CR = \frac{ConsistencyIndex(CI)}{RandomIndex(Saaty)n(RI)}, \quad (10)$$

where the components of the consistency assessment are defined in Table 14.

Table 14: Component definitions for the AHP consistency ratio equation.

Component	Type	Description
CR	Dependent variable	Consistency Ratio of the pairwise comparison matrix
CI	Dynamic variable	Consistency Index computed from the matrix principal eigenvalue
RIn	System parameter	Random Index for a matrix of size n (from Saaty's reference table)

using Saaty's Random Index table. If $CR > 0.10$, the interface flags the questionnaire as inconsistent and invites the operator to revise the most contradictory pairwise comparison before the assessment is accepted. Declared urgency is further smoothed with an exponential moving average ($\rho = 0.3$) across consecutive weekly assessments, absorbing transient stress spikes without erasing a sustained change in perception.

Figure 12: AHP questionnaire page with the consistency-ratio validation gate.

AHP is viable for U_d encoding in this context. The CR validation gate, exercised by both worked examples and automated tests, detects and rejects cognitively inconsistent expert inputs before they enter the network. Its behaviour with real operators is measured in the campaign of Section 4.8.

4.5 Stochastic Judgment Modeling: Fuzzy BWM and PROMETHEE II Network Outranking

The modules `mcda/fbwm.py` and `mcda/promethee.py` implement two complementary methods: F-BWM offers a lower-effort elicitation route than crisp AHP for weighting the six KPI blocks, and PROMETHEE II turns those weights into a network-wide, non-compensatory ranking of nodes that complements the per-node adequacy score.

F-BWM Implementation. The linguistic TFN scale maps five qualitative judgements to triangular fuzzy numbers on a 1–4.5 scale (from *equally important* to *absolutely more important*), each carrying its own maximum consistency index. The consistency ratio of the Fuzzy BWM model is formulated as:

$$CR_{\text{BWM}} = \frac{\xi^*}{\text{CI}}, \quad (11)$$

where ξ^* is the optimal consistency indicator returned by a constrained non-linear optimisation that fits the fuzzy weights to the elicited Best-to-Others and Others-to-Worst comparisons, and CI is read from the strongest linguistic judgement used. If the optimisation fails to converge, the implementation falls back to uniform weights rather than raising an error, so that a single malformed elicitation cannot crash the weekly review. Fuzzy weights are defuzzified with the Graded Mean Integration Representation (GMIR):

$$w_j = \frac{l_j + 4m_j + u_j}{6}, \quad (12)$$

then renormalised so that $\sum_j w_j = 1$. Table 15 defines the terms.

Table 15: Component definitions for the Fuzzy BWM defuzzification.

Component	Type	Description
w_j	Dependent variable	Crisp normalized weight of criterion j
l_j, m_j, u_j	Dynamic variables	Lower, modal, and upper bounds of the triangular fuzzy weight
ξ^*	Dynamic variable	Optimal consistency indicator from the fitted optimisation

PROMETHEE II Implementation. PROMETHEE II is applied in production to rank the active nodes of a project on the same six weighted urgency blocks (Table 10) that feed local real urgency, giving coordinators a network-wide, non-compensatory view of which nodes are collectively driving urgency. For each pair

of active nodes (a, b) and each block m , the preference intensity $P_m(g_m(a) - g_m(b)) \in [0, 1]$ is computed with a Type V linear preference function with an indifference threshold; the aggregated preference index and net outranking flow follow the standard PROMETHEE II formulation:

$$\pi(a, b) = \sum_m \omega_m P_m(g_m(a) - g_m(b)), \quad \phi(a) = \phi^+(a) - \phi^-(a), \quad (13)$$

where $\phi^+(a)$ and $\phi^-(a)$ are the average preference of a over, respectively by, all other active nodes. Table 16 defines the outranking flow terms.

Table 16: Component definitions for the PROMETHEE II net flow.

Component	Type	Description
$\pi(a, b)$	Dependent variable	Aggregated preference index of a over b
$\phi(a)$	Dependent variable	Net outranking flow representing overall preference ($\in [-1, 1]$)
$\phi^+(a), \phi^-(a)$	Dynamic variables	Positive (dominance) and negative (dominated) outranking flows

The same weights ω_m used in the local urgency aggregation (Equation 4) are reused as the PROMETHEE criteria weights, so that the network ranking and the per-node urgency score remain mutually consistent rather than relying on a separately calibrated weight set. Both modules are unit tested and wired into the production ranking path (`services/orchestrator.py`).

4.6 Weekly Review Cycle and Calibrated Event Engine

The weekly review cycle is what keeps the digital twin’s KPI state current without free-form manual data entry: every write is structured, signed, and auditable. This section describes the cycle itself, then the calibrated event engine that translates real-world disruptions into KPI updates.

The Weekly Cycle. The system is operated as a serious game advancing in discrete weeks (`core/clock.py` implements `SystemClock` for live use, `GameClock` for week-by-week advancement, and `FixedClock` for deterministic testing). At each new week, every active node is flagged as due for review, and the operator completes four sequential sections (`services/weekly.py`, served by the `/hebdo` page, Figure 14):

1. the AHP questionnaire, pre-filled with the previous week's values, with a one-click "unchanged" confirmation;
2. the week's KPI values, shown against the previous week for quick differencing;
3. progress on active milestones, compared against theoretical planning progress;
4. any supply chain events that occurred during the week.

All four sections are written as a single signed, atomic transaction, so that no partial or anonymous review can corrupt the node's history.

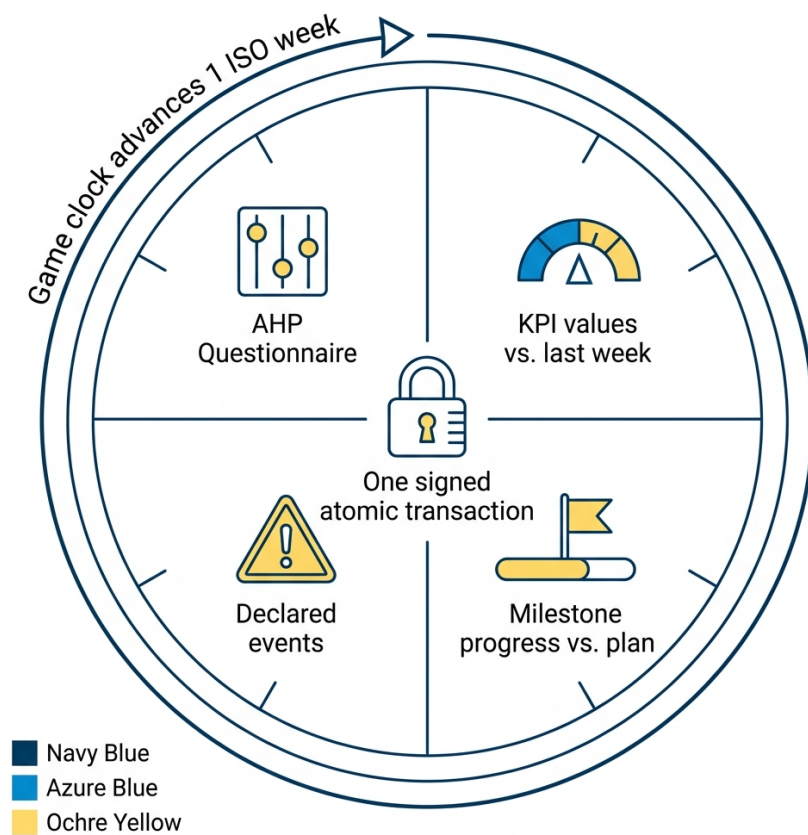


Figure 13: The four-section weekly review cycle committed as a single signed transaction.

Figure 14: Weekly review page: the four sequential sections (AHP, KPIs, milestones, events).

Calibrated Event Engine. Rather than asking operators to manually recompute the KPI impact of a disruption, the system exposes a taxonomy of fourteen calibrated event types (`domain/events.py`), listed in Table 17, each mapped to one or more KPI updates through one of three calibration operators.

Table 17: The fourteen calibrated event types.

panne_machine	retard_fournisseur	greve
hausse_tarif	rupture_matiere	accident
non_conformite_qualite	perturbation_transport	instabilite_politique
cyber_incident	hausse_energie	alerte_financiere_fournisseur
perte_capacite	pic_demande	

The three calibration operators are: a Beta-Bernoulli Bayesian update for failure probabilities,

$$p_{\text{post}} = \text{clip}\left(\frac{p \times n_0 + k}{n_0 + n}, 10^{-4}, 0.99\right), \quad (14)$$

with a prior weight $n_0 = 26$ (roughly six months of weekly observations); an exponential moving average for continuous variables such as OEE, $x_{\text{new}} = (1 - \lambda)x_{\text{old}} + \lambda x_{\text{obs}}$ with $\lambda \in [0.3, 0.5]$; and a ratchet (max) update for severity, $s_{\text{new}} = \max(s_{\text{old}}, s_{\text{event}})$. Table 18 defines the shared terms.

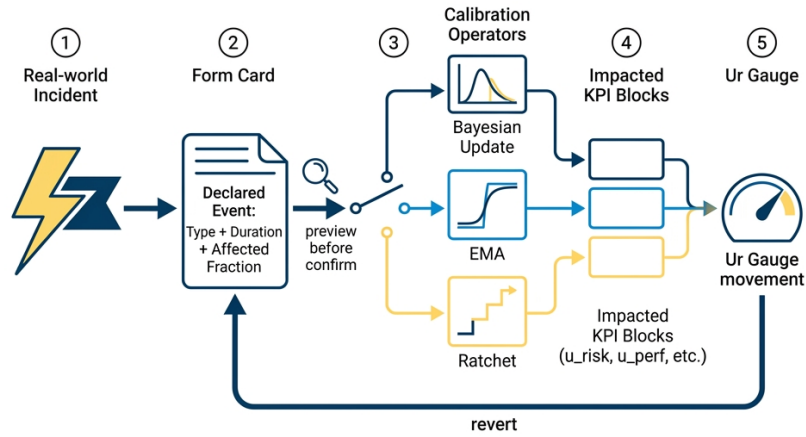


Figure 15: From a declared event to a calibrated KPI update: the three calibration operators.

Table 18: Component definitions for the event calibration operators.

Component	Type	Description
p_{post}	Dependent variable	Revised failure probability after the event
n_0	System parameter	Prior equivalent sample size (26, roughly six months)
(k, n)	Event input	Equivalent failures/trials implied by the declared event's severity
x_{new}	Dependent variable	Revised continuous KPI (e.g., availability)
λ	System parameter	EMA reactivity weight, $\in [0.3, 0.5]$

Numerical Example: Strike Event. A node declares a greve (strike) lasting $d = 24$ hours affecting a fraction $p_f = 0.50$ of the workforce. Equivalent lost hours

over a 168-hour week: $H_{\text{lost}} = d \times p_f = 12$, so $f_{\text{lost}} = 12/168 \approx 0.0714$ and the observed availability is $A_{\text{obs}} = 1 - 0.0714 \approx 0.9286$. Updating a baseline availability $A_{\text{old}} = 0.95$ with EMA reactivity $\lambda = 0.5$: $A_{\text{new}} = 0.5(0.95) + 0.5(0.9286) \approx 0.9393$. Risk severity is updated in ratchet mode: $S_{\text{new}} = \max(0.30, 0.50) = 0.50$. Operators can preview the transient impact of a declared event before confirming it, and reverting an event recomputes the baseline KPIs automatically.

The weekly cycle and event engine let coordinators keep the digital twin's KPI state current from structured, auditable inputs rather than free-form edits. This is what makes the resulting urgency history usable as a calibration dataset for the campaign of Section 4.8.

4.7 Explainability, Export, and Operational Hardening

When the digital twin flags a node, the coordinator's first question is *why*. The last three capabilities of the demonstrator exist to answer that question exactly, to let the data leave the tool, and to make sure nothing is ever lost.

Explainability. The per-node explanation page (`services/explain.py`) answers "why this score?" with exact decompositions, not approximations. Declared urgency splits into one additive contribution per AHP criterion:

$$\kappa_j = \frac{w_j (s_j - 1)}{8}, \quad \sum_j \kappa_j = U_d, \quad (15)$$

so the page can state, for instance, that half of a node's declared urgency comes from the time-window criterion. Local real urgency splits the same way through the logarithm of Equation 4: each block contributes $\ell_m = -\omega_m \ln(1 - u_m)$, the block shares $c_m = \ell_m / \sum_k \ell_k$ sum to one, and $U_{r,\text{local}} = 1 - \exp(-\sum_m \ell_m)$ reconstructs the score exactly. A coordinator can therefore see which block drives a node's urgency, and whether the gap between U_d and U_r is panic or hidden danger (Figure 16).

SupplyScore
Projets
Onboarding
Hébergement
Questionnaire
Dashboard
Simulation
Édition
Pondération
Graphique
Rapport
Admin
Opérateur
Activer
Annuler

Pondération des critères d'urgence (FBWM)

À quoi servent ces poids ?
FBWM pondère les 6 BLOCS KPI de l'urgence réelle Ux — l'ANP du questionnaire reste dédié aux 4 critères du besoin déclaré U6.
 Seulement 2n-3 comparaisons (2 pour 6 blocs) suffisent, là où une matrice complète en requerrait 15.
 Poids actuels du projet : Temps 0.285 - Capacité 0.091 - Performance (DEI) 0.156 - Risque 0.229 - Coût 0.164 - CO2 0.075.

1. Meilleur et Pire critère
 Le FBWM ancre toutes les comparaisons sur ces deux extrêmes.
 Critère le PLUS important Critère le MOINS important
 Choisir un bloc... Choisir un bloc...

2. Jugements linguistiques (2n-3)
 Choisissez d'abord le critère le PLUS important et le MOINS important.
 0/9 jugements validés

3. Aperçu des poids résolus
 L'aperçu se recalcule à chaque jugement modifié — rien n'est persisté.
 Choisissez deux critères distincts (Meilleur et Pire) puis les 2n-3 jugements.

4. Enregistrement (projet actif)
 L'enregistrement enregistre vos jugements instantanés (CR < 0.10), rattache les poids à la semaine T00 du projet et réévalue tous les scores.
 Enregistrer les poids pour ce projet Revenir aux poids uniformes

Figure 16: Criteria weighting interface (Fuzzy-BWM) feeding the explainability decomposition.

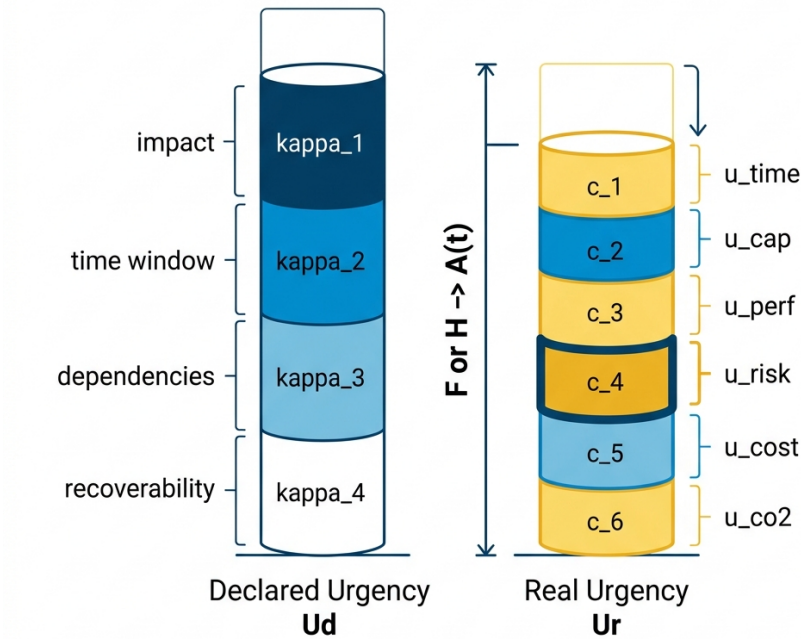


Figure 17: Per-node explainability: exact additive decomposition of U_d and U_r .

Export. One action compiles the global registry and the per-node history into a twelve-worksheet Excel workbook (`services/exports.py`): project metadata, nodes, arcs, milestones, KPI history, urgency history, assessments, weekly reviews, KPI snapshots, events, decisions, and two audit-log sheets. All timestamps are converted to the coordinator's local timezone. Analyses can therefore run outside the tool, on frozen data, which the validation campaign relies on.

Operational Hardening. The persistence layer is SQLite, so data safety is handled explicitly (`data/backup.py`): Write-Ahead Logging is enabled, an integrity check runs before every backup, and the native `sqlite3.backup` API copies each database without locking the running application. Automatic backups run every 24 hours with a retention cap of 20 archives, and schema migrations are versioned (Figure 18).

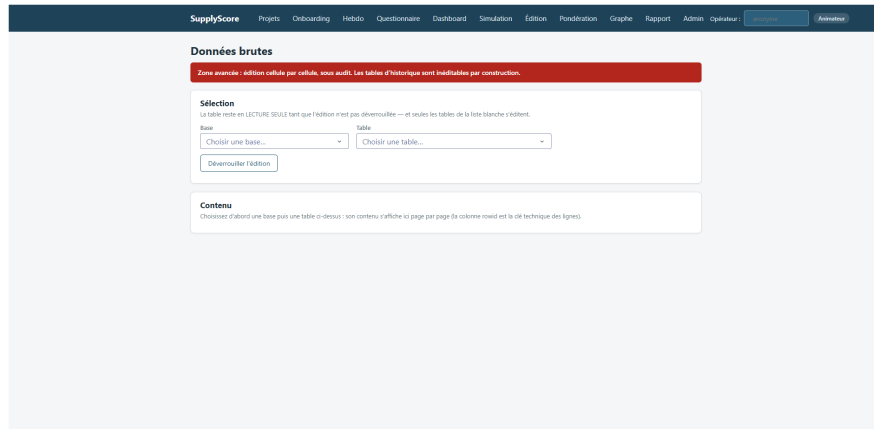


Figure 18: Administration page: database sizes, migration version, and backup controls.

These three capabilities are not peripheral. They are what make the adequacy score auditable and the digital twin operable outside a development environment, and they are preconditions for the campaign below.

4.8 Serious-Game Validation Campaign: Replaying the Semiconductor Crisis

Everything up to this point establishes that the model is implemented, internally consistent, and tested. What it does not establish is whether the adequacy signal says anything true about real operators facing a real crisis. This final section presents the validation campaign built for that purpose: a serious-game replay of the 2020–2022 semiconductor shortage, designed, calibrated through five dry runs, and pre-registered before execution.

Campaign Design. The campaign, presented to participants under the cover story *Programme HÉLIOS*, replays the semiconductor crisis over 18 weekly game turns compressed into nine working days (two turns per day, morning and late afternoon). The supply network has eight nodes anchored on the real actor types of the crisis: a satellite prime, an avionics integrator, an electronics manufacturing services plant, a component distributor, a freight operator, a semiconductor foundry at the origin of the shortage, a second foundry, and a raw-materials supplier. One consultant plays each node.

The two urgency signals come from different worlds, which is the point of the design. Real urgency U_r is fed from public time series of the actual crisis (INSEE

production and price indices, WSTS worldwide semiconductor billings), frozen in a data pack whose SHA-256 hashes were recorded before the first turn. Declared urgency U_d comes from the consultants through the same weekly AHP questionnaire described earlier, under ten minutes per turn, with the consistency gate active.

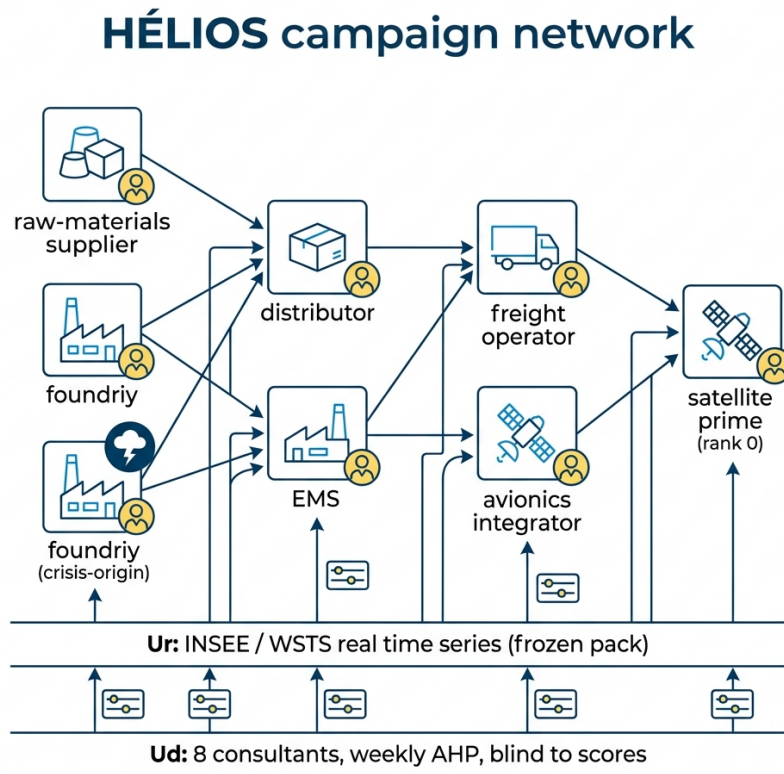


Figure 19: The eight-node HÉLIOS campaign network and its two data sources.

Protocol Integrity. The experimental protocol was frozen before the first real turn. Consultants see only their own role sheet and the questionnaire page; dashboards and scores stay hidden until the debrief. No discussion between roles is allowed during the campaign. Two placebo sheets are injected at pre-defined turns to control for suggestibility, and every deviation from protocol is logged the same day in a campaign journal. The hypothesis register is the one of Section 2.4: H1 to H6, each with a numeric threshold fixed in advance, so that success and failure are defined before the data exists. Any analysis outside that register is labelled exploratory.

Five dry runs with synthetic operator profiles calibrated the scenario before freezing. They also produced a methodological finding in their own right: when the whole network saturates at the peak of the crisis, the criticality index degenerates (realised risk can no longer be shocked, Section 4.3.4), so saturated turns are excluded from the H3 test and the criticality comparison is restricted to deep-tier suppliers.

Empirical Calibration of the Adequacy Signal. The campaign feeds the calibration service (`services/calibration.py`), which addresses Challenge 3 (Section 2.5) and hypotheses H2 and H4: does the hidden-risk signal H have predictive value beyond a raw KPI threshold? For each recorded (node, week) pair, the historical hidden risk is compared against negative outcomes observed in a subsequent four-week window (a missed milestone, an abandoned node, or a critical event). The resulting confusion matrix yields precision, recall, and F1:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (16)$$

The service flags any calibration run with fewer than 30 observation points as too small for statistically meaningful conclusions. The campaign is sized precisely to clear that bar: 8 nodes over 18 turns yield 144 (node, week) observations (Figure 20).

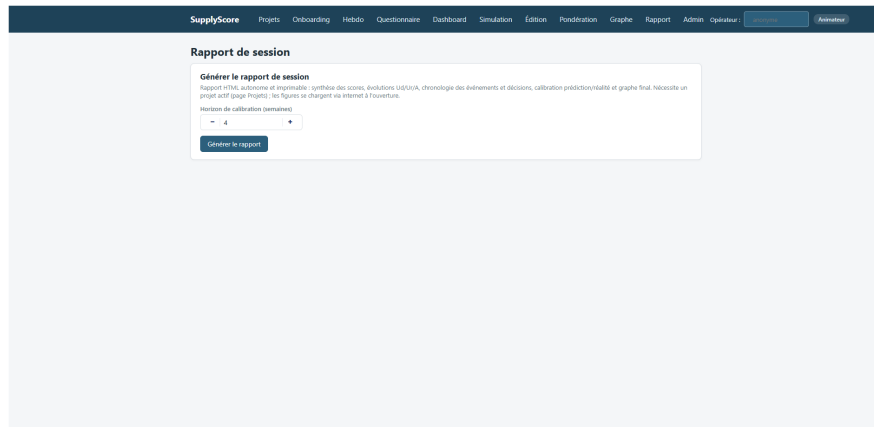


Figure 20: Calibration report: confusion matrix and precision/recall/F1 for the hidden-risk signal.

[TO COMPLETE AFTER THE CAMPAIGN: results per hypothesis. For each of H1–H6, report the measured value against its pre-registered threshold (lagged correlation per node for H1; the turn at which $H > 0.3$ on the crisis-origin node for H2;

the criticality rank per informative turn for H3; the full confusion matrix with precision/recall/F1 at $H \geq 0.5$ for H4; the declared-urgency delta of untouched nodes on press-event turns for H5; the U_d vs. U_r decay slopes after the peak for H6) and state verdict: supported / refuted / inconclusive.]

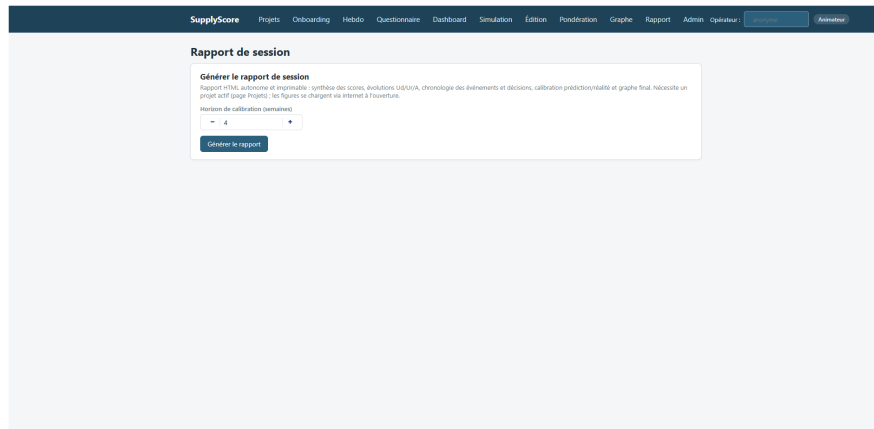


Figure 21: Declared vs. real urgency trajectories measured during the HÉLIOS campaign.

[TO COMPLETE AFTER THE CAMPAIGN: lessons learned. What the debrief revealed (including the reveal question: at which turn did each consultant recognise the real crisis?), which model assumptions held, which parameters need recalibration (λ_{under} , α , block weights ω_m), and what changes before a client-facing deployment. The next steps drawn from these lessons close the report in Section 5.]

5 Conclusions

This R&D operation delivered *SupplyScore*: a multi-block, noisy-OR real urgency model propagated across the supply chain DAG, an asymmetric cognitive adequacy score confronting declared and computed urgency, and a persistent, multi-tenant digital twin operated as a weekly-cadence serious game, with a pre-registered validation campaign (Section 4.8) sized to measure the predictive validity of the hidden-risk signal. The scope of the delivered framework was bounded by explicit, documented design decisions rather than left implicit. This section states those limitations directly, draws the next steps, and closes with a personal assessment of the internship.

5.1 Limitations

Table 19 lists the limitations judged material enough to state directly, together with their operational implication.

Table 19: Documented limitations and their operational implications.

Limitation	Implication
No probabilistic guarantee	The framework produces probabilities, scores, and confidence intervals, not certainties; a model remains a simplification of the underlying reality.
Score freshness follows input cadence	Without an ERP connector, a node's score is never more current than its last submitted weekly review; manual entry is an accepted trade-off, not an oversight.
AHP is limited to four criteria	Saaty's method degrades past roughly seven pairwise-compared criteria, which is why declared urgency stays on four criteria while the larger set of KPI blocks is weighted through Fuzzy-BWM instead.
Event reversibility is conditional	Reverting a declared event is only guaranteed correct if no other write has touched the same KPI in the meantime; otherwise the system raises an explicit conflict for manual resolution rather than silently rebasing later writes, which would be mathematically dishonest.
Event calibration priors are defensible, not measured	The Beta-Bernoulli prior weight ($n_0 = 26$) and EMA reactivity coefficients follow standard conjugate Bayesian revision but are not empirically fitted; the serious game itself is the intended instrument for recalibrating them.
Empirical validation is designed, not yet executed	The calibration service and its confusion-matrix methodology are implemented and tested, but at the time of writing the HÉLIOS campaign (Section 4.8) has not yet been run with its human participants; no predictive-validity claim is made until it has.
Local time urgency looks only at the next milestone	A distant milestone that is silently drifting stays invisible to u_{time} until it becomes the next active one; the underlying function already receives the full milestone list, so a weighted multi-milestone extension is possible without a model change.
Double counting is possible	An operator can both declare a calibrated event and manually correct the same KPI in the same week; the interface warns but does not forbid this, since the human operator remains the final authority.

5.2 Next Steps

[TO COMPLETE AFTER THE HÉLIOS CAMPAIGN: the next steps of the programme, drawn from the campaign's lessons learned (Section 4.8). Expected content: which hypotheses the campaign supported or refuted and what that implies; the recalibration of the governance parameters (λ_{under} , α , ω_m , n_0) from measured data; and the decision points for the following phase (multi-expert elicitation, coupling with the DISCO platform, adequacy-driven scheduling).]

5.3 Bilan Personnel et Compétences Mobilisées

5.3.1 Compétences techniques mobilisées

Cette opération de R&D a mobilisé des compétences dans quatre domaines distincts, dont l'articulation constitue l'essentiel de la difficulté technique du sujet :

- **Aide à la décision multi-critères** : élicitation par comparaisons par paires (AHP, Fuzzy-BWM), validation de cohérence, agrégation par surclassement (PROMETHEE II).
- **Modélisation probabiliste** : agrégation noisy-OR, simulation de Monte Carlo, révision bayésienne conjuguée (Beta-Bernoulli) en régime de données rares.
- **Algorithmique de graphes** : propagation sur graphe orienté acyclique multi-rangs, tri topologique, simulation de choc et calcul de criticité réseau.
- **Génie logiciel** : architecture d'un système multi-tenant en Python, persistance SQLite durcie (WAL, sauvegarde, migrations), interface web Dash, démarche qualité en jalons avec couverture de tests bloquante et tests de bout en bout automatisés.

5.3.2 Apprentissages et difficultés rencontrées

- Quelle a été la principale difficulté technique ou méthodologique rencontrée, et comment a-t-elle été surmontée ?
- Qu'est-ce qui aurait été fait différemment avec le recul ?
- Quelles compétences (techniques ou non) ont le plus progressé au cours du stage ?
- En quoi ce stage éclaire-t-il un projet professionnel ?

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Glossary

- $A(t)$ Adequacy score at time t , quantifying the asymmetric Prospect Theory penalty between U_d and $U_r(t)$: $A = 100 \times (\exp(-\text{penalty}) - \text{floor}) / (1 - \text{floor})$ with $\text{penalty} = \lambda_{\text{under}} H^\alpha + \lambda_{\text{over}} F^\alpha, \in [0, 100]$. 77
- F, H False urgency $F = [U_d - U_r]_+$ (unjustified panic) and hidden risk $H = [U_r - U_d]_+$ (silent danger), the two asymmetrically penalised pathologies underlying $A(t)$. 77
- U_d Declared urgency, a scalar $\in [0, 1]$ elicited from the human operator via an AHP or Fuzzy-BWM questionnaire, propagated downstream across the supply chain DAG through the leaky coefficients γ_{ik} . 77
- $U_r(t)$ Real (operationally computable) urgency at time t : a weighted probabilistic OR aggregation of six local KPI blocks (time, capacity, performance, risk, cost, CO2), $U_{r,\text{local}} = 1 - \prod_m (1 - u_m)^{\omega_m}$, propagated upstream across the supply chain DAG through the leaky coefficients β_{ji} . 77
- $\Delta U(t)$ Raw urgency mismatch: $U_d - U_r(t) \in (-1, 1)$; positive = over-triage (false urgency), negative = under-triage (hidden risk). 77
- $\Delta U_{r,\text{final}}$ Network criticality delta: the increase in real urgency propagated to final-customer nodes when a given node's local urgency is forced to saturation in a systematic shock simulation. 77
- ω_m Aggregation weight of KPI block m in the real urgency noisy-OR aggregation, reused unchanged as the PROMETHEE II criteria weight for network-wide node outranking. 77

Acronyms

AHP Analytic Hierarchy Process. 31, 77

BST Bilan Scientifique et Technique. 77

BWM Best-Worst Method. 31, 77

CI Consistency Index. 51, 52, 77

CR Consistency Ratio. 31, 51, 52, 53, 77

CSRD Corporate Sustainability Reporting Directive. 77

DAG Directed Acyclic Graph. 77

DIN Direction Innovation Numérique. 9, 77

DISCO Supply Chain Illustrator (Digital Twin, DIN ALTEN). 77

DT Digital Twin. 29, 30, 77

F-BWM Fuzzy Best-Worst Method. 31, 77

GLEC Global Logistics Emissions Council. 9, 33, 77

HGV Heavy Goods Vehicle. 11, 77

IoT Internet of Things. 9, 77

IPG Itinerary Performance Index. 11, 77

KPI Key Performance Indicator. 77

LPPL Log-Periodic Power Law. 25, 77

MCDM Multi-Criteria Decision-Making. 21, 28, 30, 77

MTS Manchester Triage System. 32, 77

PI Physical Internet. 21, 33, 77

PROMETHEE Preference Ranking Organisation METHod for Enrichment Evaluation. 77

RI Random Index (Saaty). 51, 52, 77

SC Supply Chain. 29, 30, 77

TFN Triangular Fuzzy Number. 31, 77

TMT Temporal Motivational Theory. 26, 77

TTR Time To Recovery. 11, 77

TTS Time To Service. 11, 77

VECTO Vehicle Energy Consumption calculation TOol. 33, 77

XAI Explainable Artificial Intelligence. 77